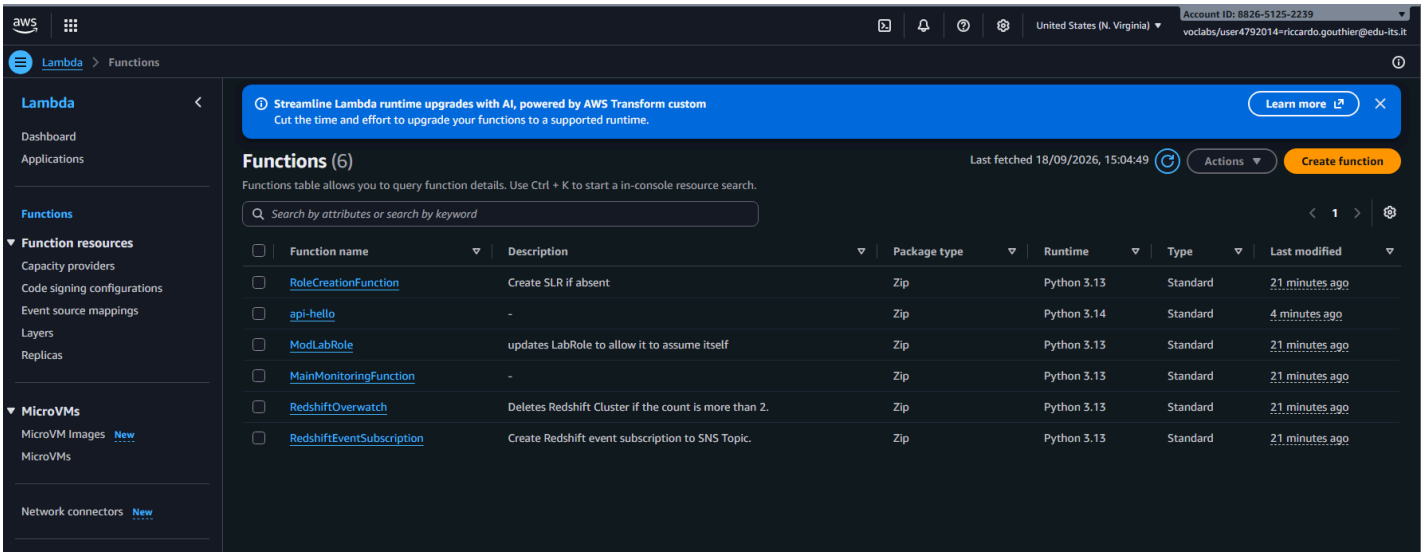
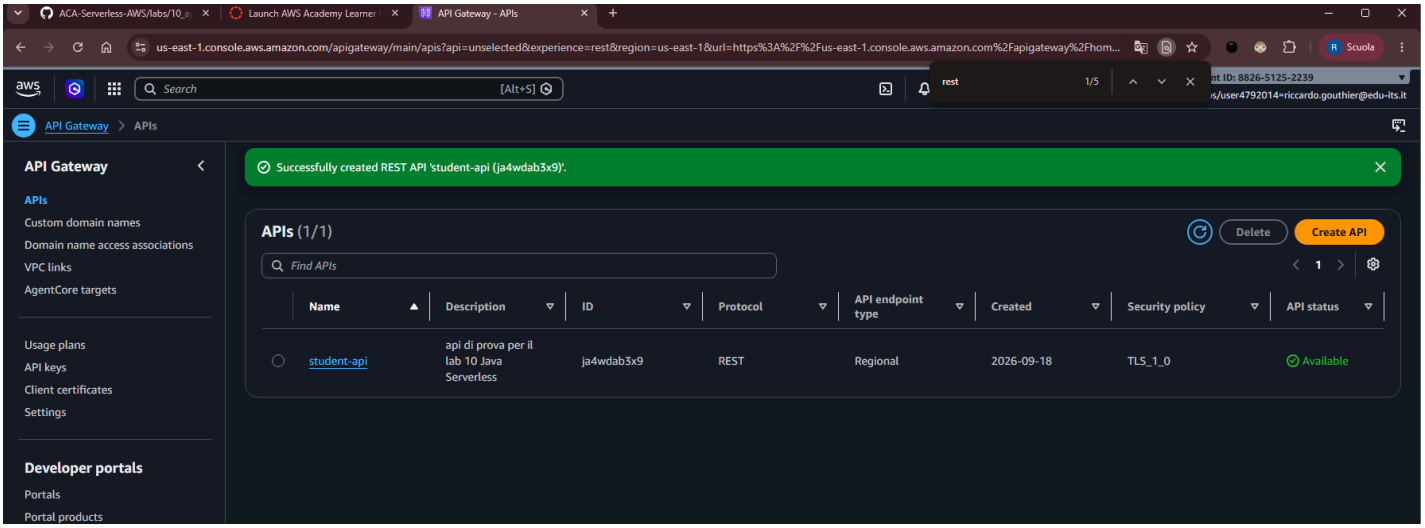
Lab 10

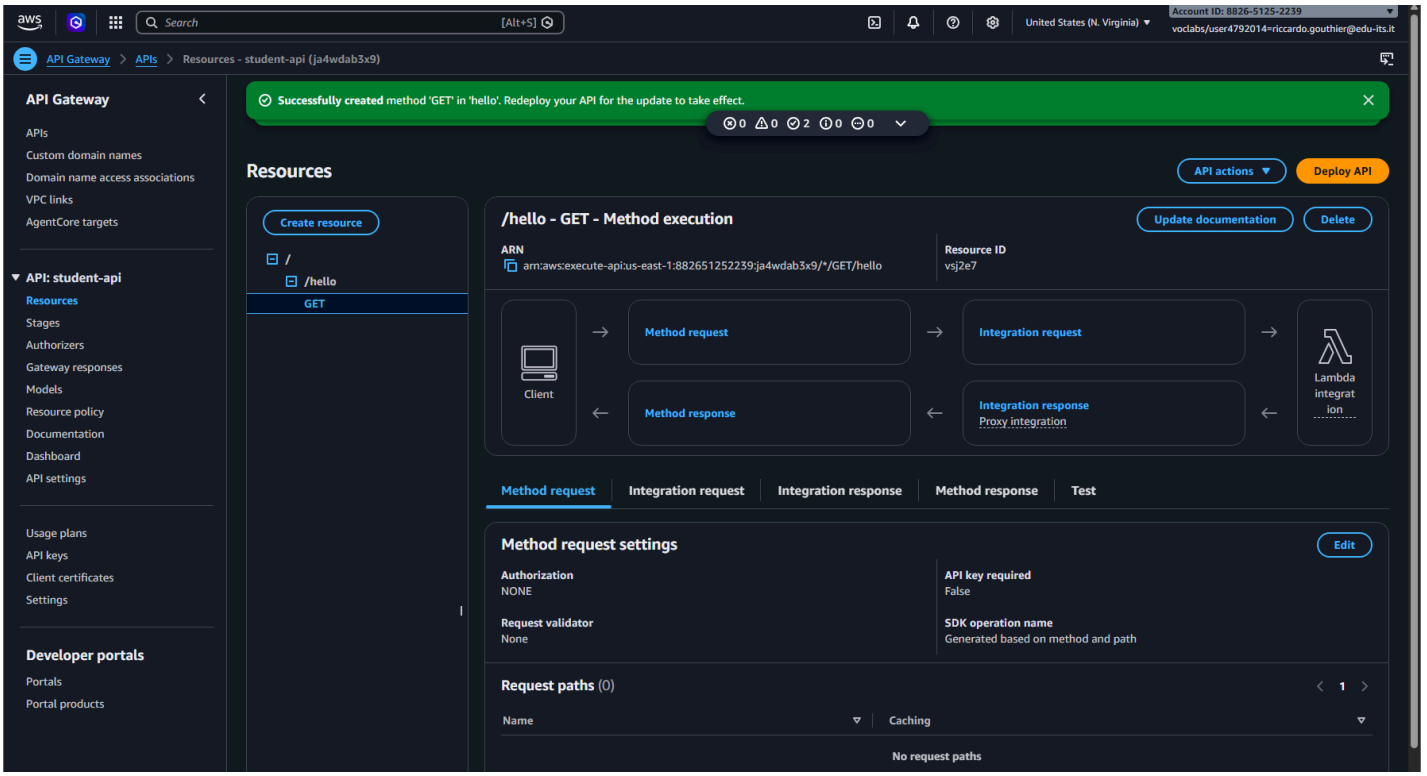
creazione Lambda



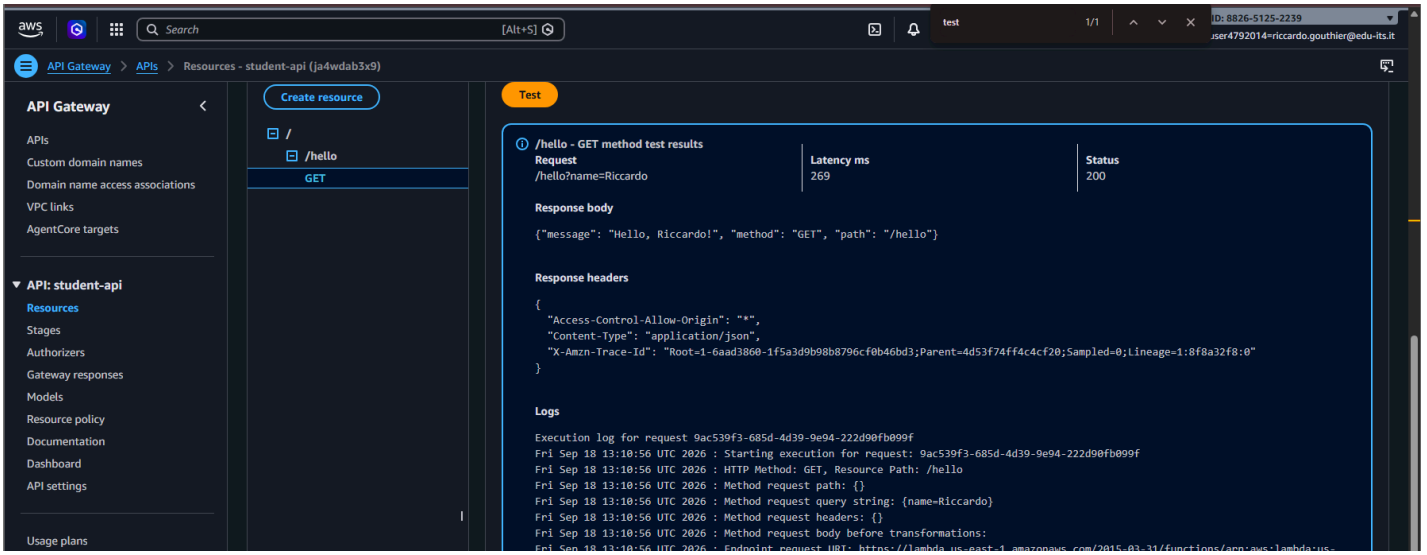
creazione API



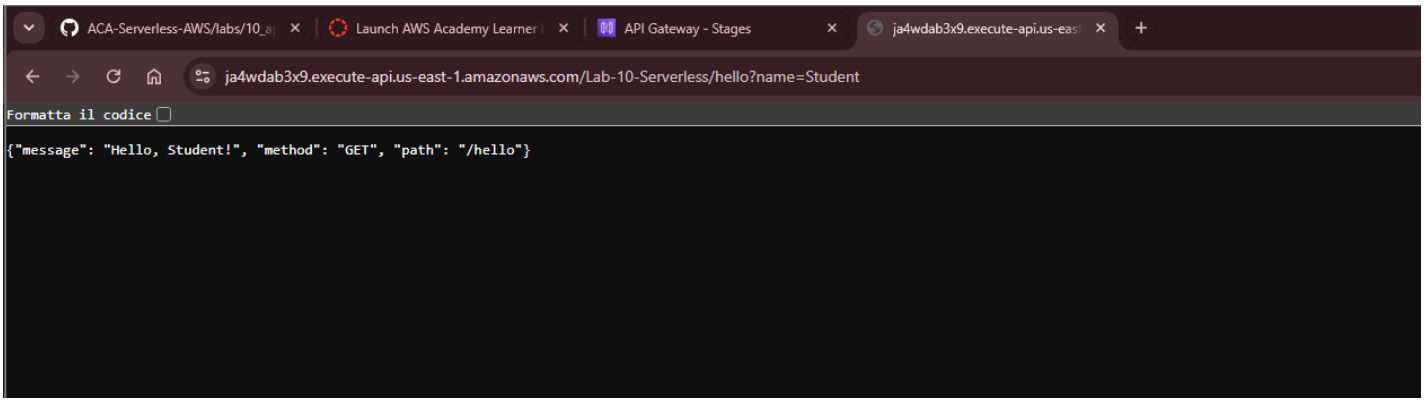
risorsa /hello + metodo GET



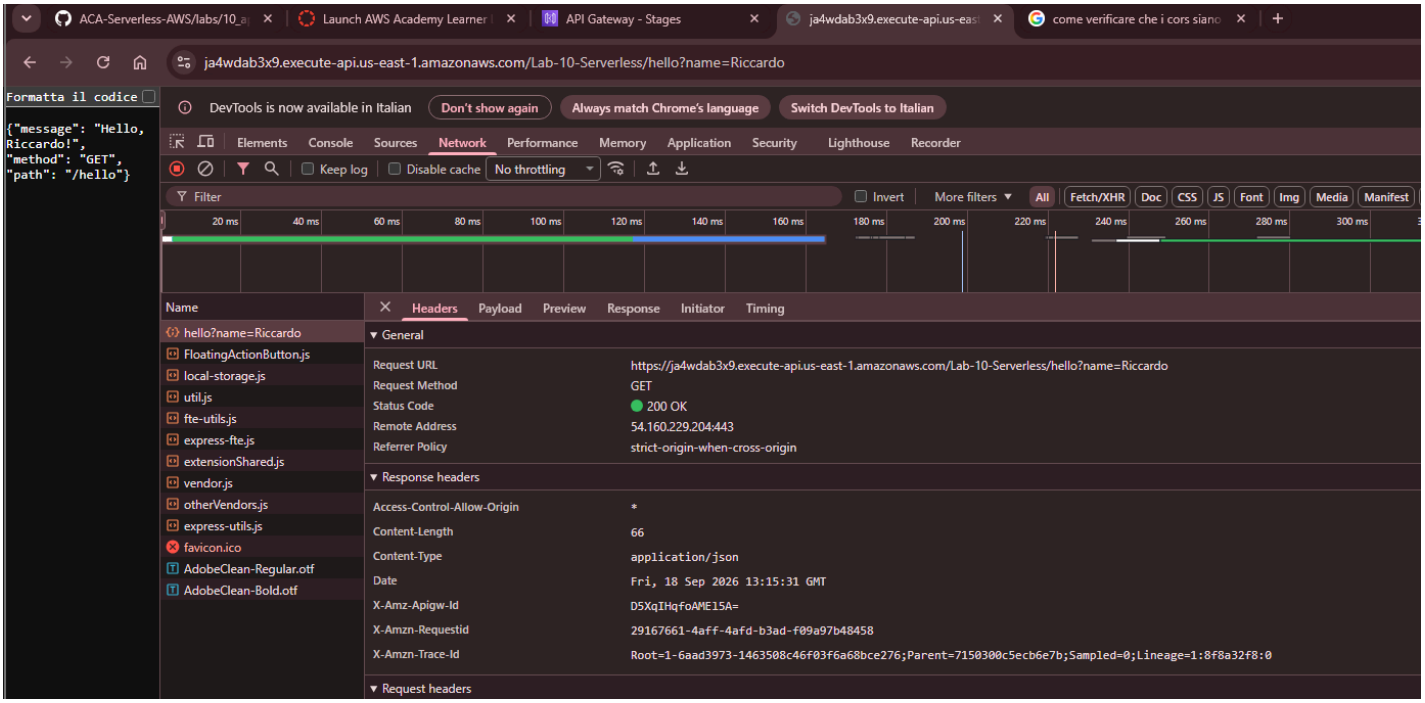
test in console



deploy API

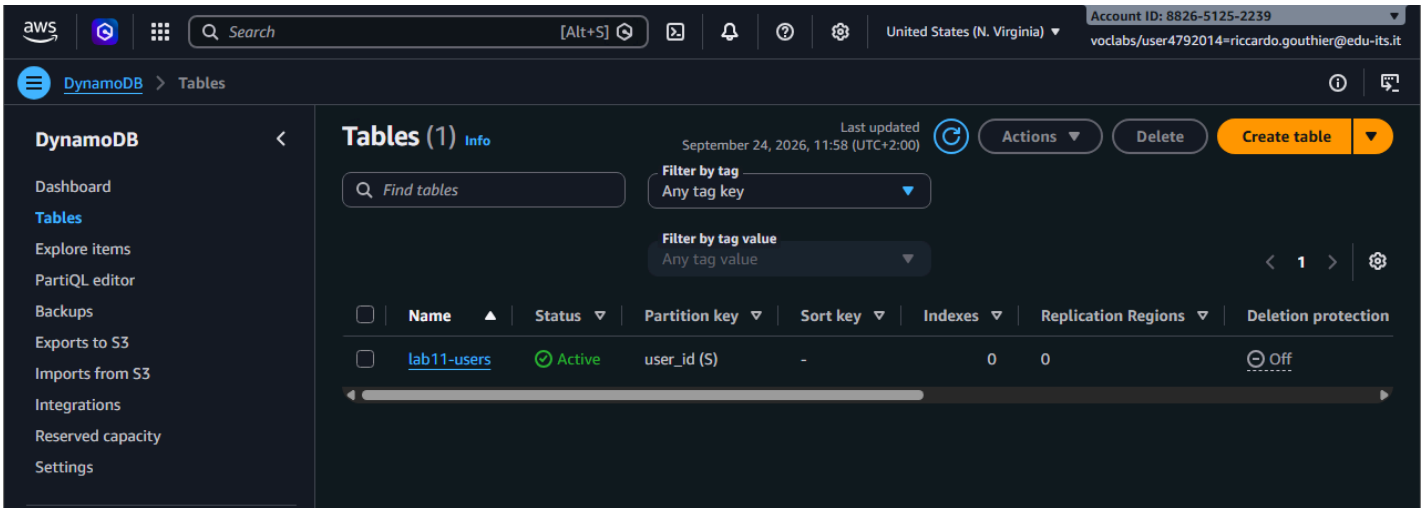


CORS

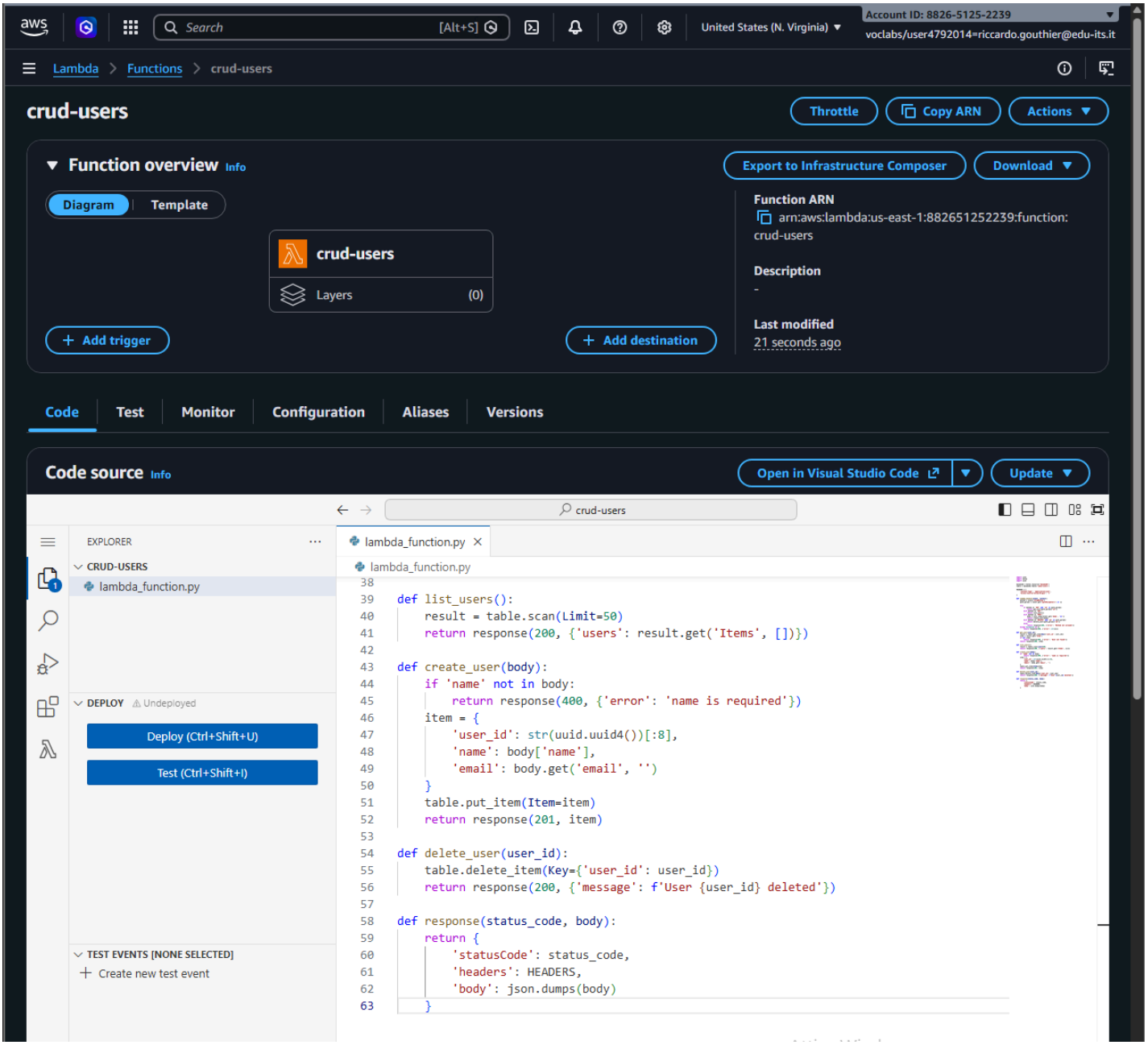


Lab 11

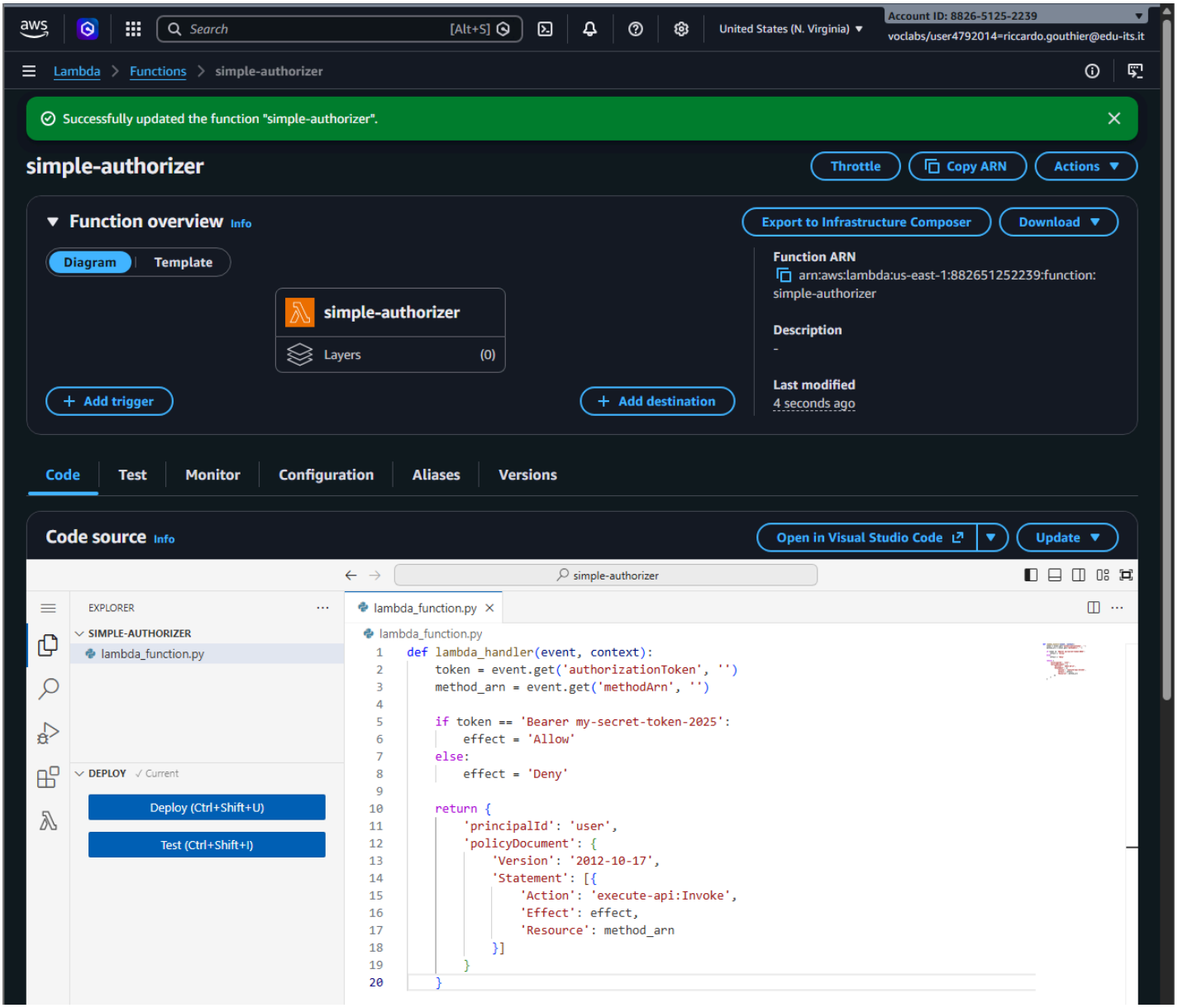
Creare tabella DynamoDB



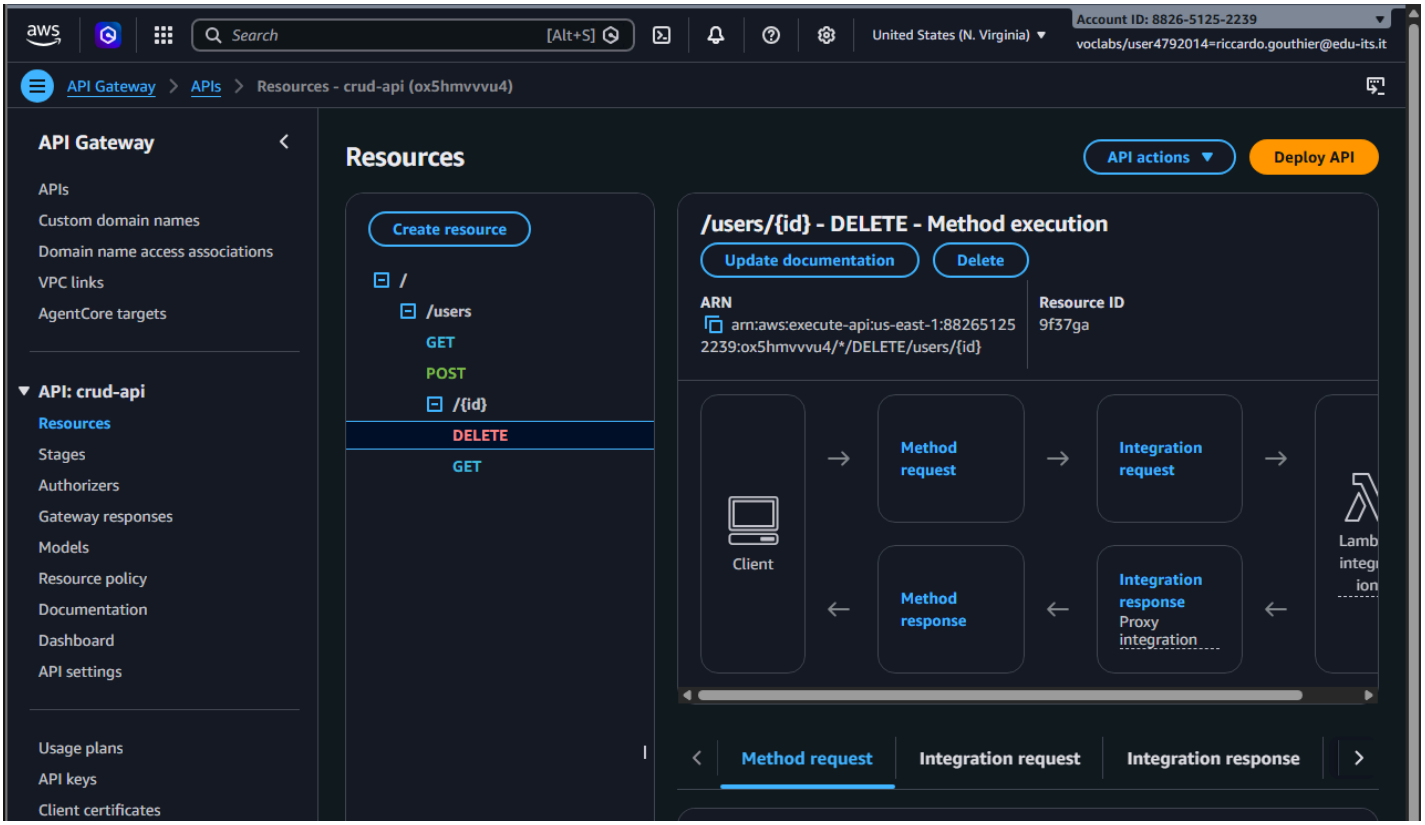
Creare funzione CRUD



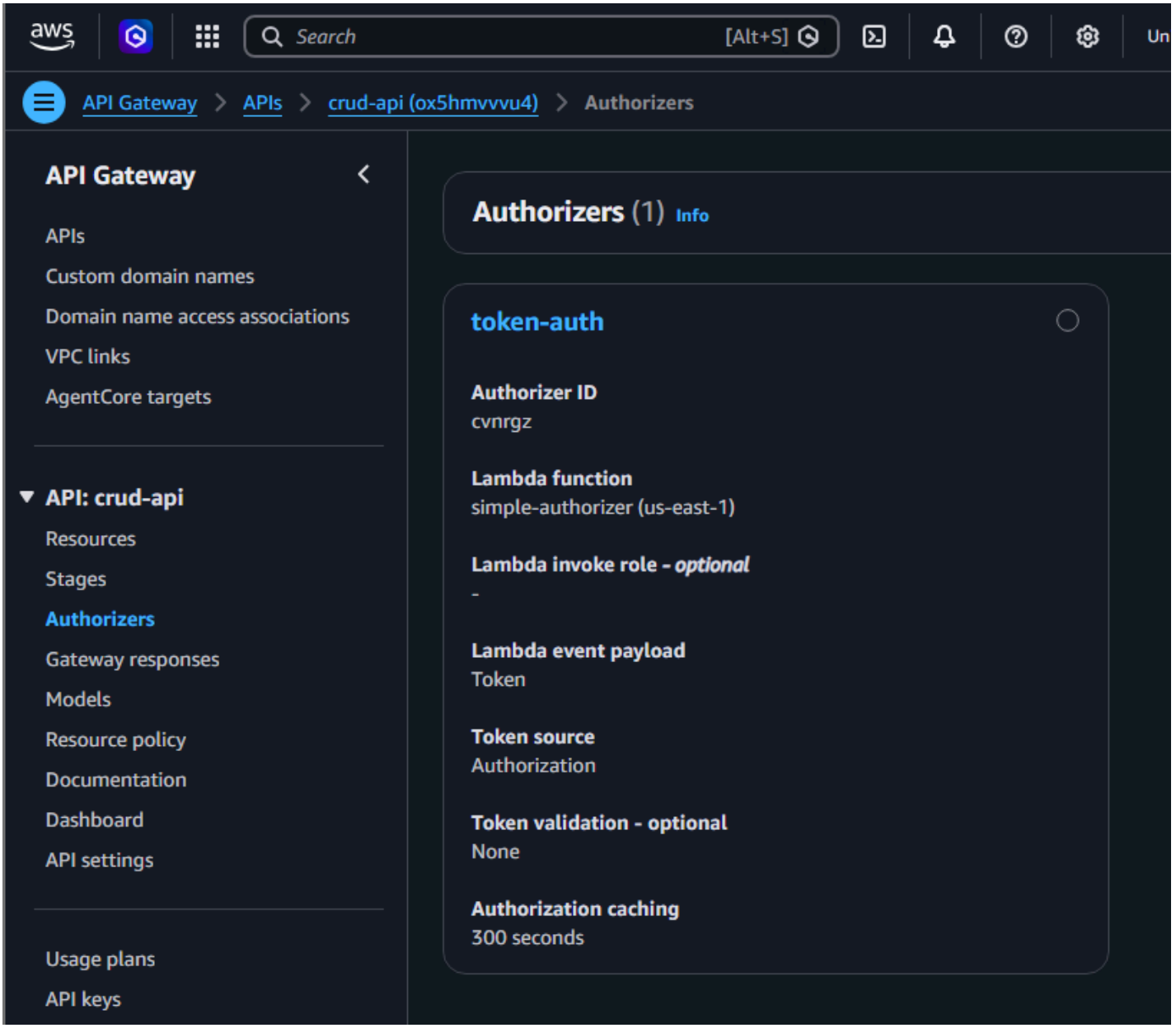
Creare Lambda authorizer



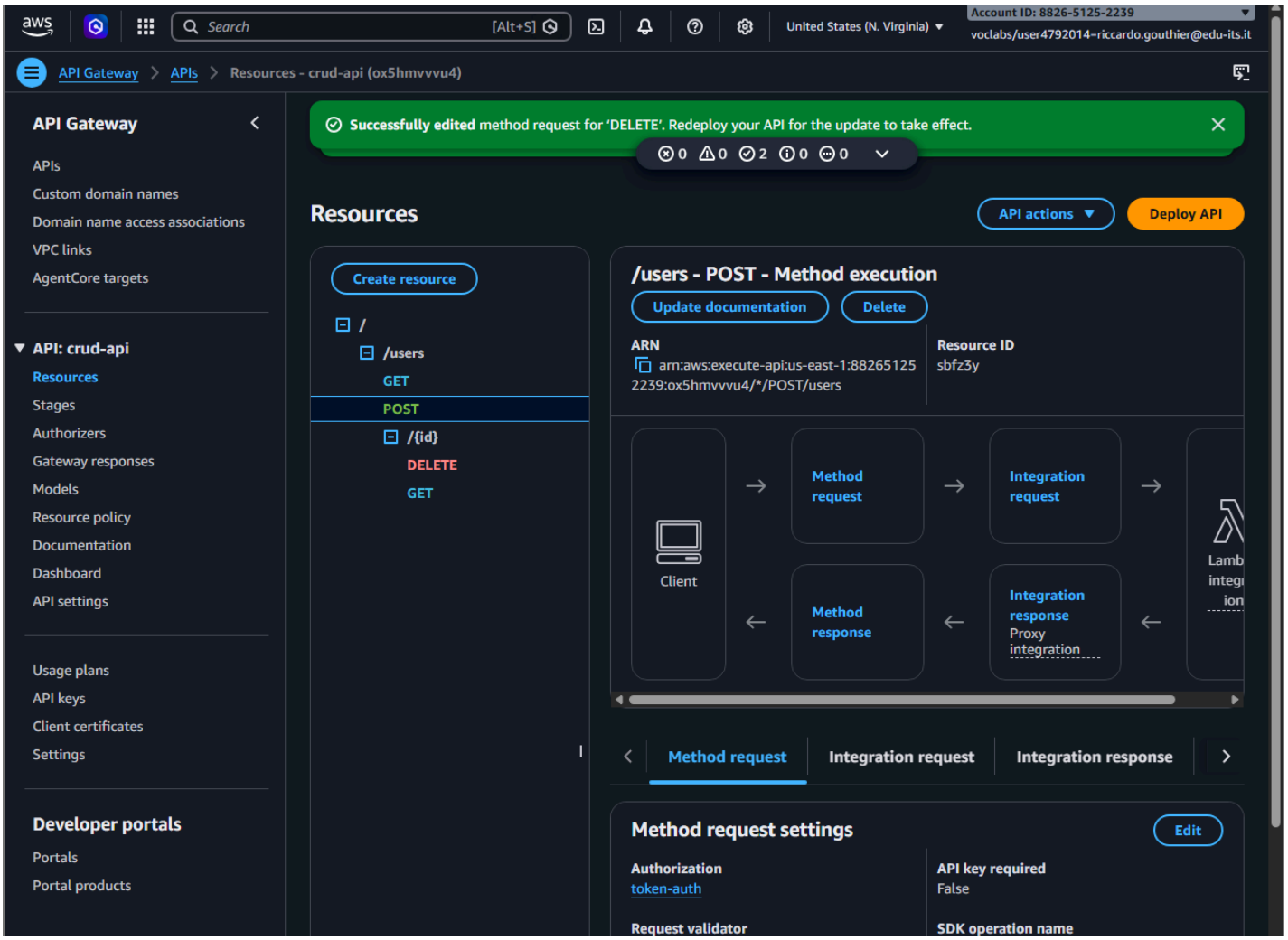
Creare API Gateway



Configurare authorizer

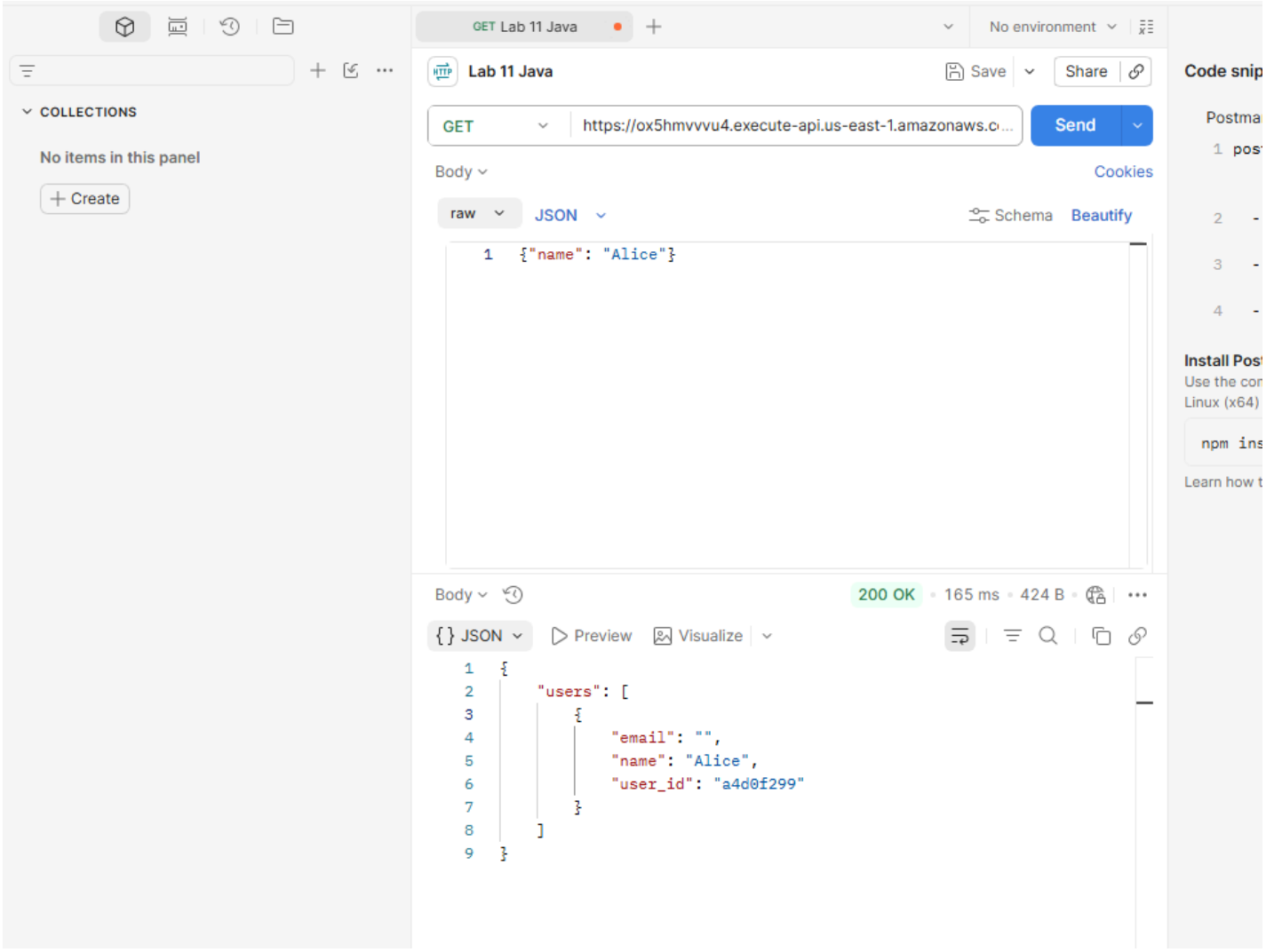


Step opzionale
Nota: chiamate POST/DELETE senza header Authorization: Bearer my-secret-token-2025 restituiscono 401/403.



Test finale del GET dopo aver fatto POST con postman

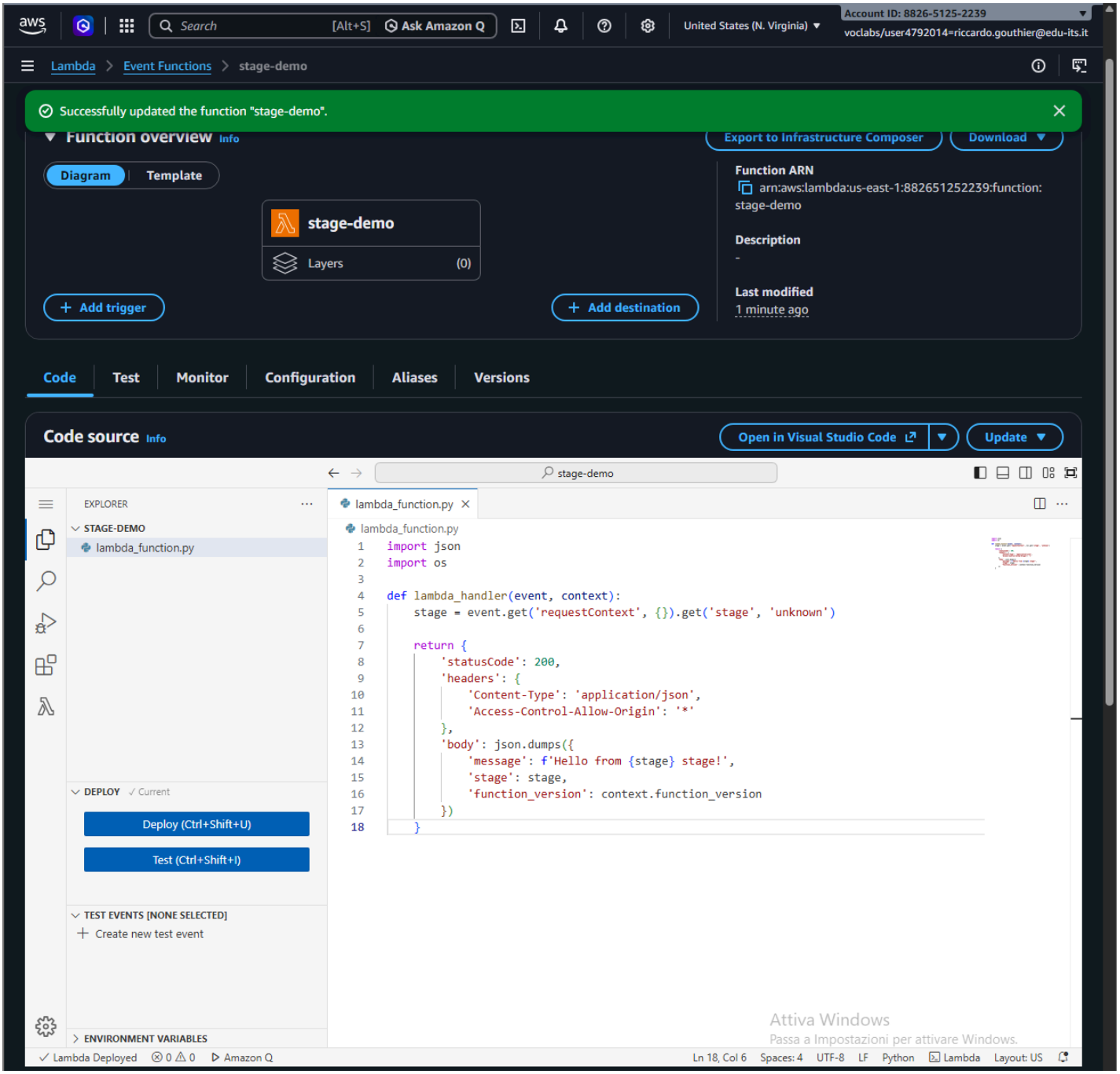
```
request 'https://ox5hmvvvu4.execute-api.us-east-1.amazonaws.com/dev/users?=' \
  --header 'Authorization: Bearer my-secret-token-2025' \
  --header 'Content-Type: application/json' \
  --body '{"name": "Alice"}'
```



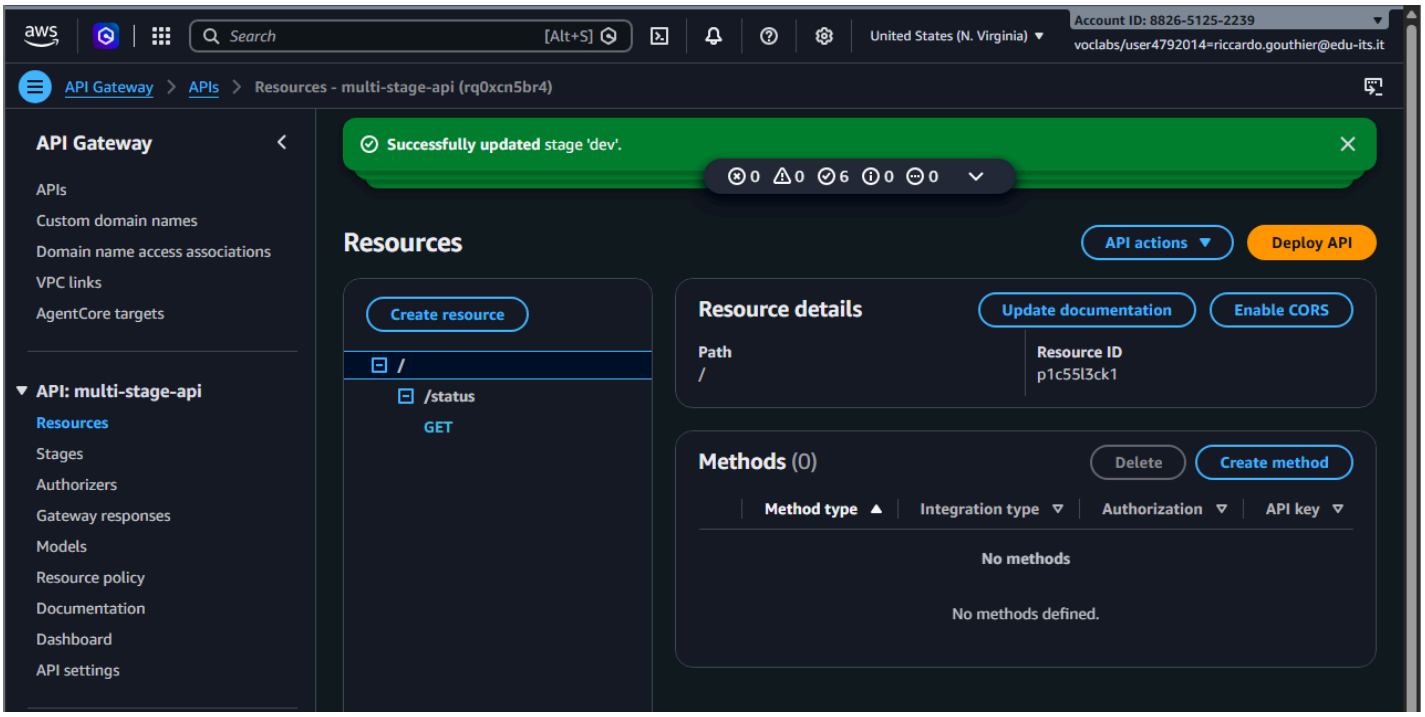
Lab 12

Parte A - Versioni, Stage e Monitoraggio API

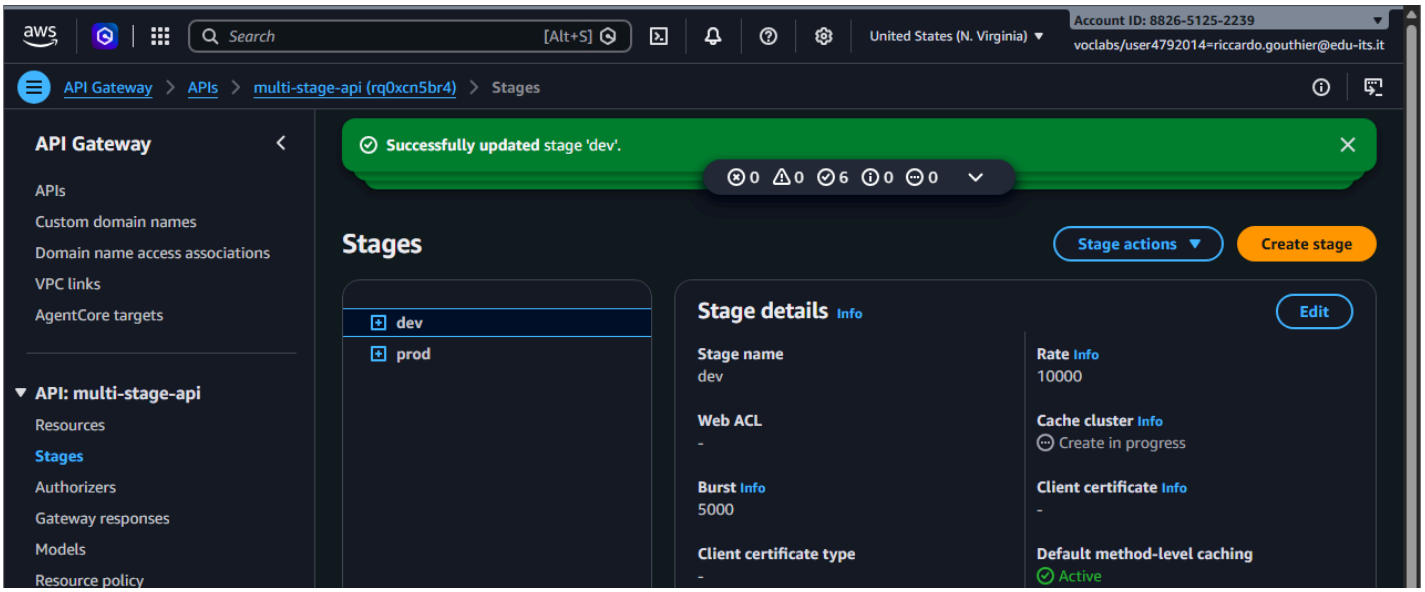
Creare la funzione Lambda



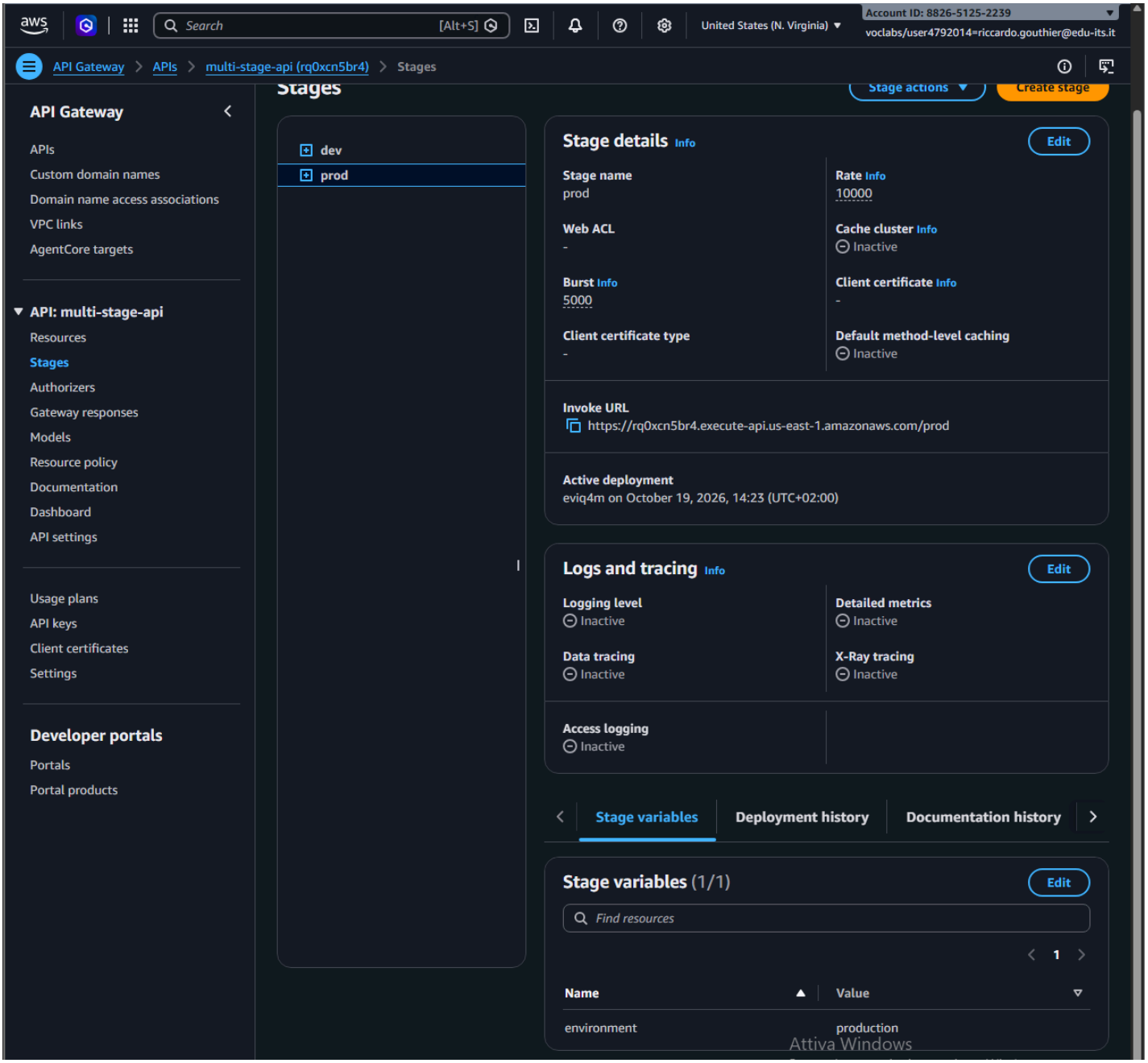
Creare l'API con stage multipli



Creati dev e prod con “Default method-level caching: Active”

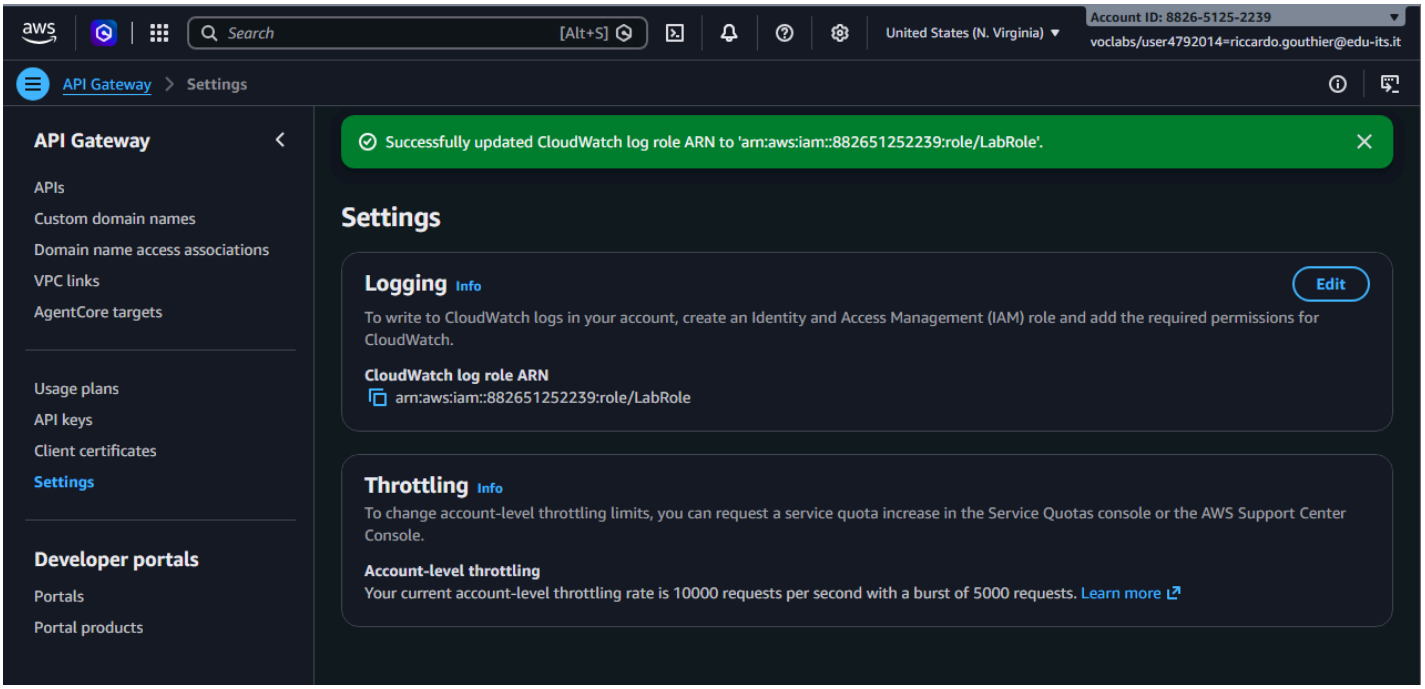


Configurare stage variables

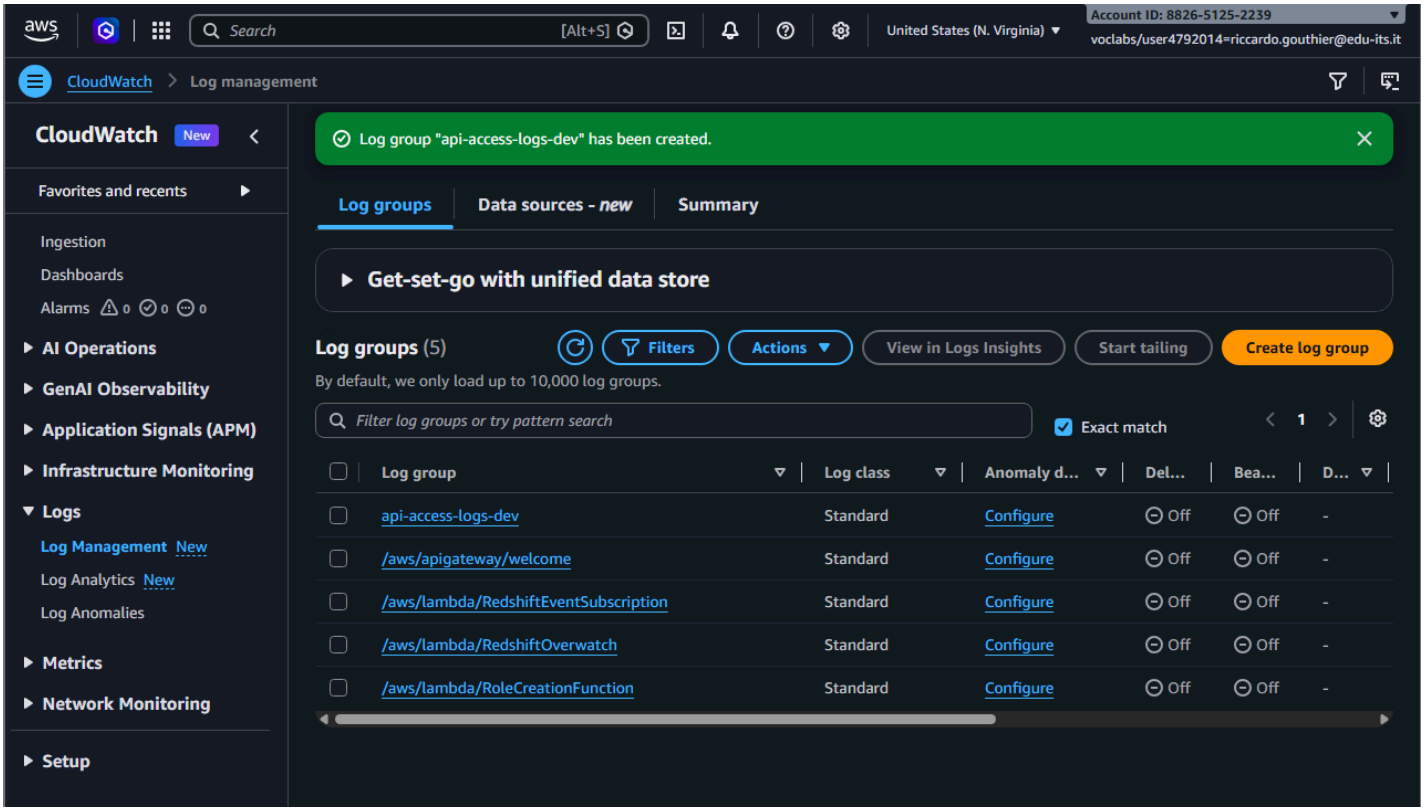


(è mostrato solo il tab prod ma la stage variable “environment” è stata fatta anche per lo stage dev)

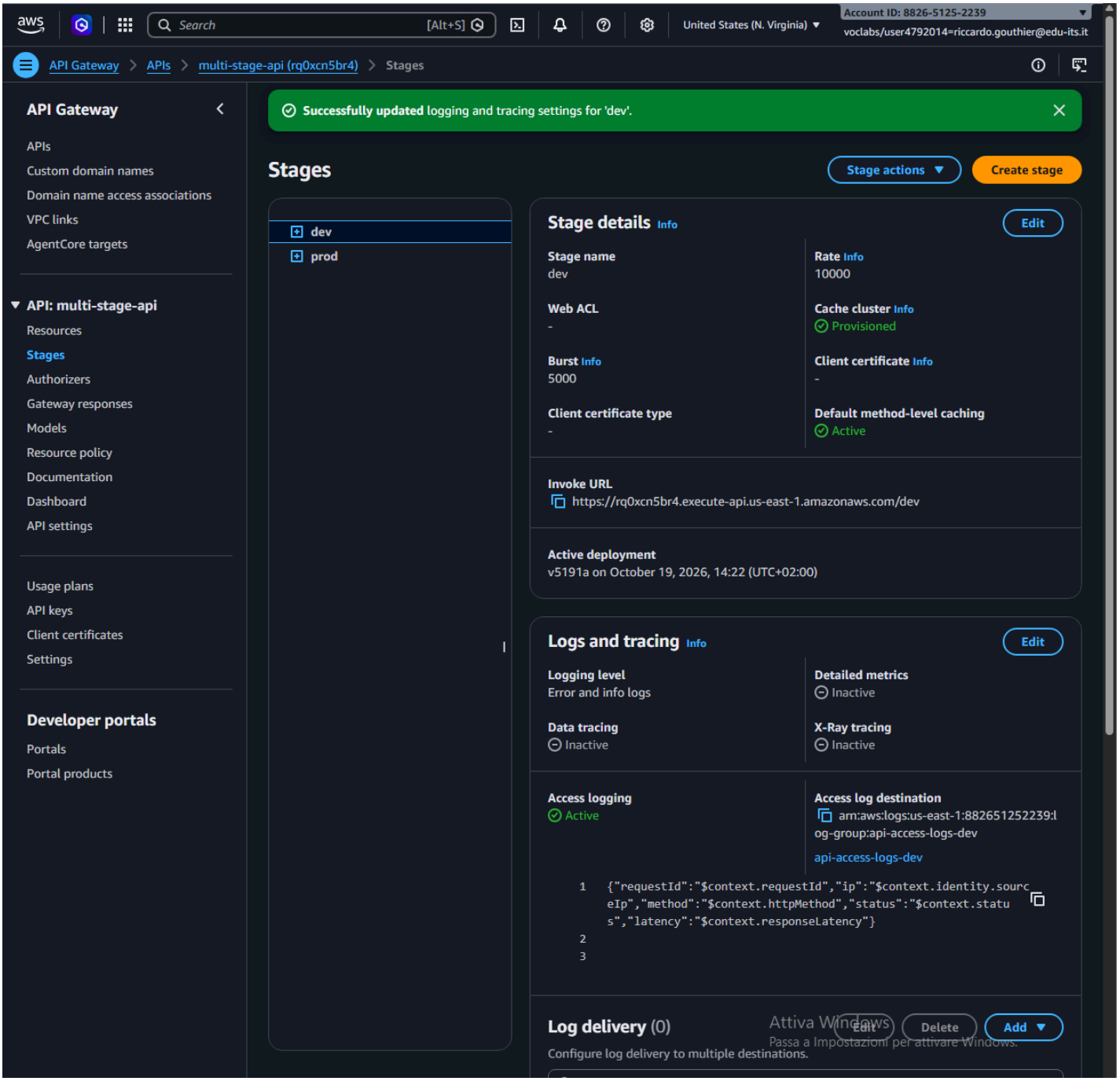
Abilitare i log di accesso
Ruolo CloudWatch


Search [Alt+S]
United States (N. Virginia)
Account ID: 8826-5125-2239
voclabs/user4792014=riccardo.gouthier@edu-its.it
API Gateway > Settings
API Gateway
APIs
Custom domain names
Domain name access associations
VPC links
AgentCore targets
Usage plans
API keys
Client certificates
Settings
Developer portals
Portals
Portal products
Successfully updated CloudWatch log role ARN to 'arn:aws:iam::882651252239:role/LabRole'.
Settings
Logging Info
Edit
To write to CloudWatch logs in your account, create an Identity and Access Management (IAM) role and add the required permissions for CloudWatch.
CloudWatch log role ARN
arn:aws:iam::882651252239:role/LabRole
Throttling Info
To change account-level throttling limits, you can request a service quota increase in the Service Quotas console or the AWS Support Center Console.
Account-level throttling
Your current account-level throttling rate is 10000 requests per second with a burst of 5000 requests. Learn more

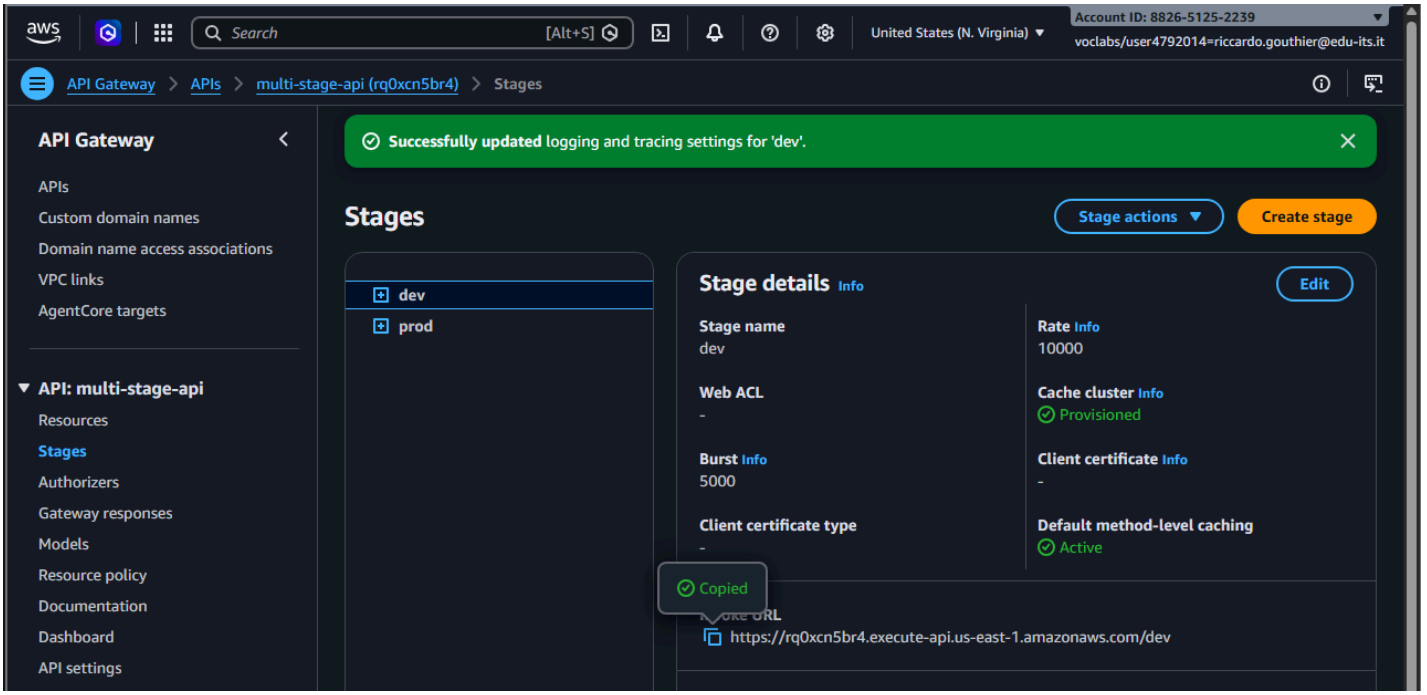
Log group

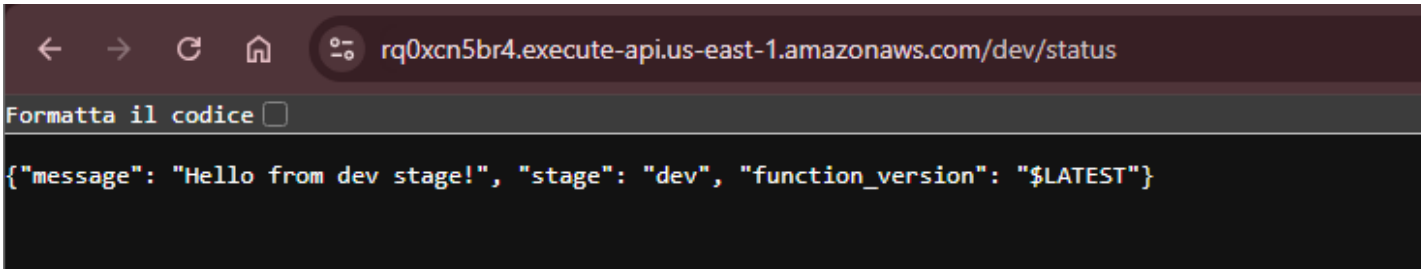

Search [Alt+S]
United States (N. Virginia)
Account ID: 8826-5125-2239
voclabs/user4792014=riccardo.gouthier@edu-its.it
CloudWatch > Log management
CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management New
Log Analytics New
Log Anomalies
Metrics
Network Monitoring
Setup
Log group "api-access-logs-dev" has been created.
Log groups
Data sources - new
Summary
Get-set-go with unified data store
Log groups (5)
Filters
Actions
View in Logs Insights
Start tailing
Create log group
By default, we only load up to 10,000 log groups.
Filter log groups or try pattern search
Exact match
Log group
Log class
Anomaly d...
Del...
Bea...
D...
api-access-logs-dev Standard Configure Off Off -
/aws/apigateway/welcome Standard Configure Off Off -
/aws/lambda/RedshiftEventSubscription Standard Configure Off Off -
/aws/lambda/RedshiftOverwatch Standard Configure Off Off -
/aws/lambda/RoleCreationFunction Standard Configure Off Off -

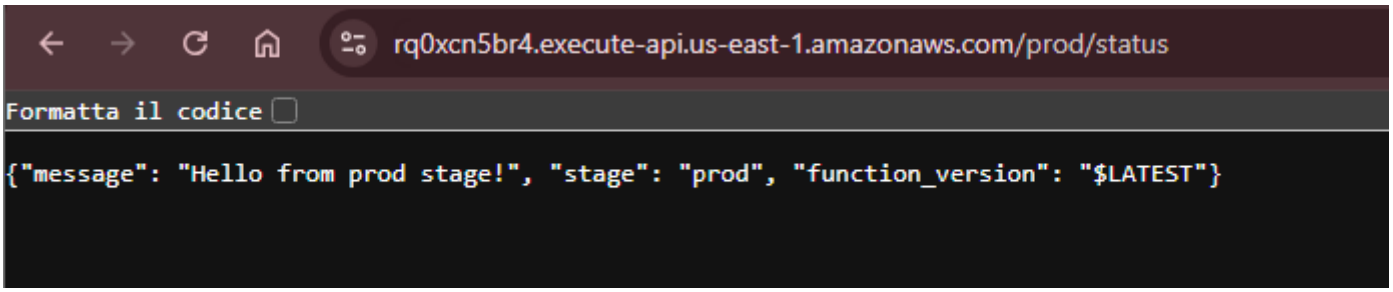
Config stage



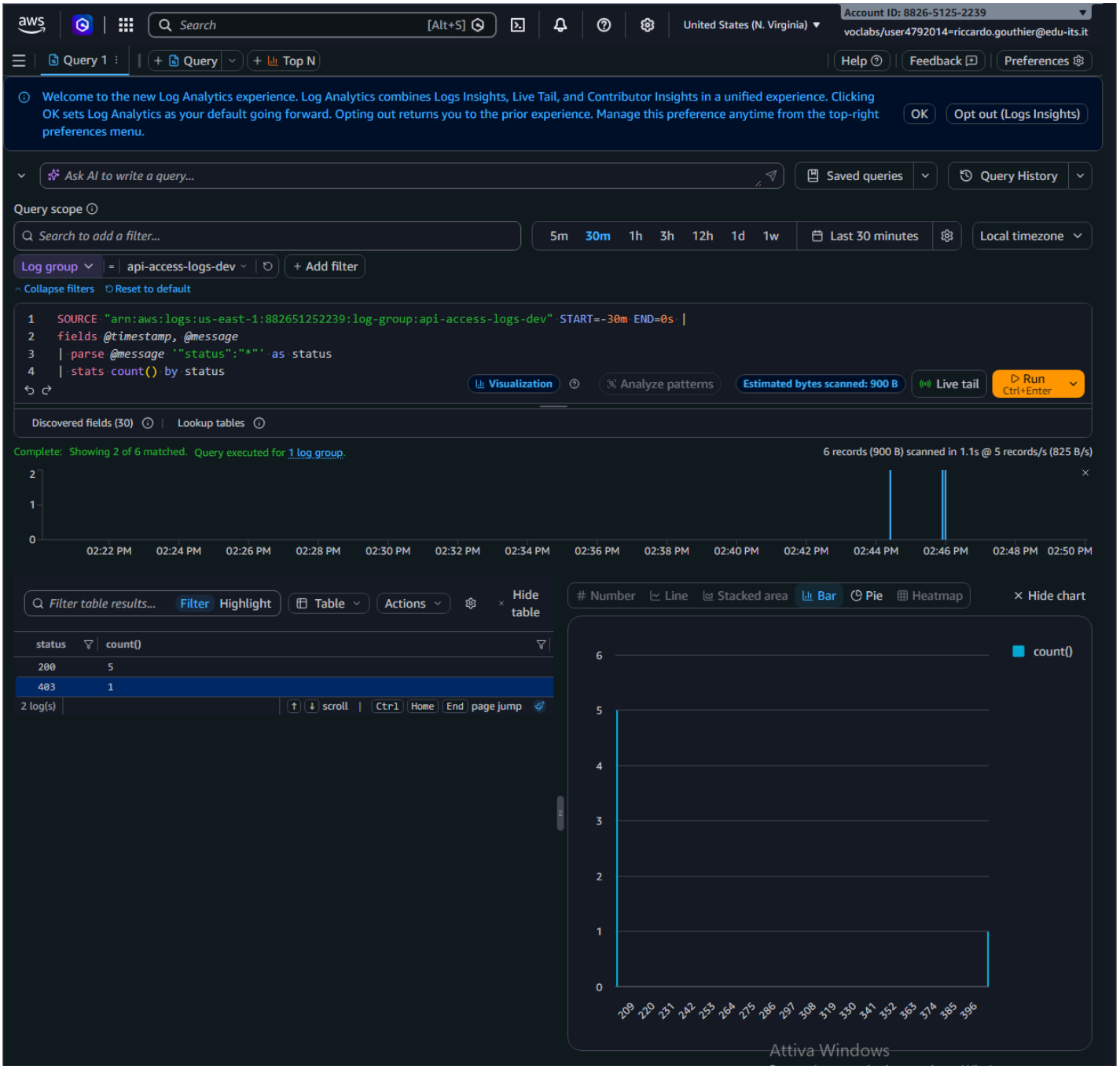
Testare e generare traffico





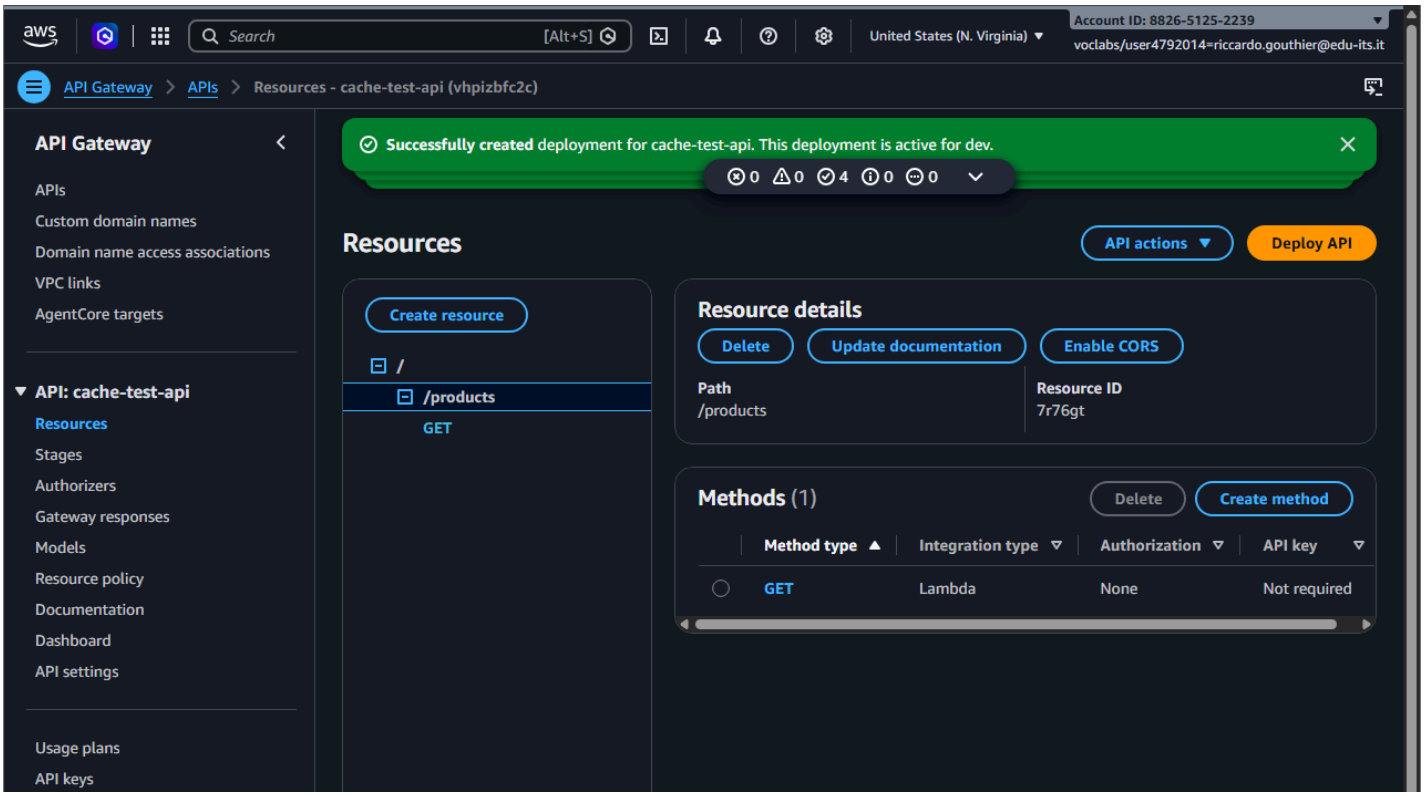


Analizzare i log

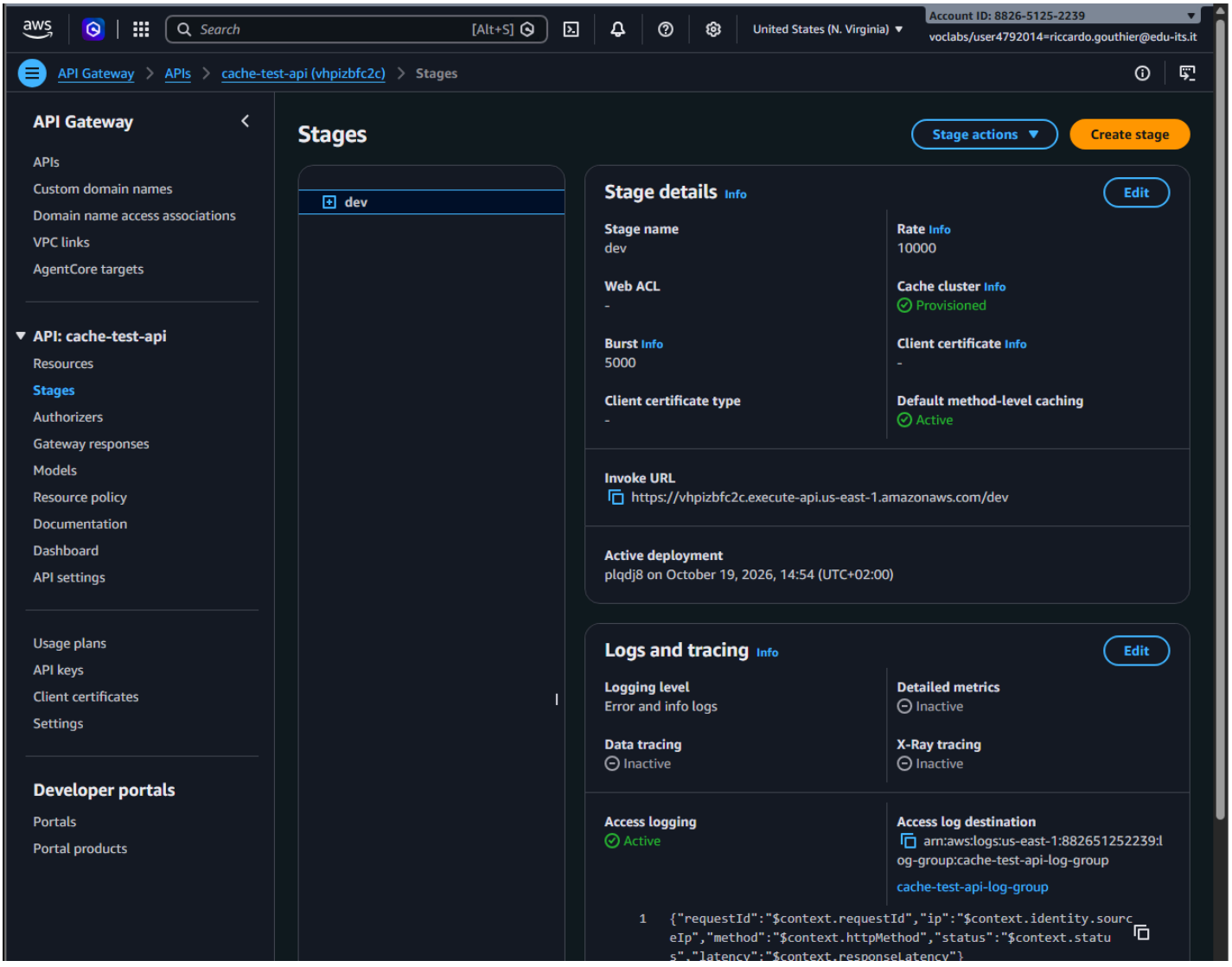


Parte B - Caching, Monitoraggio Avanzato e Revisione

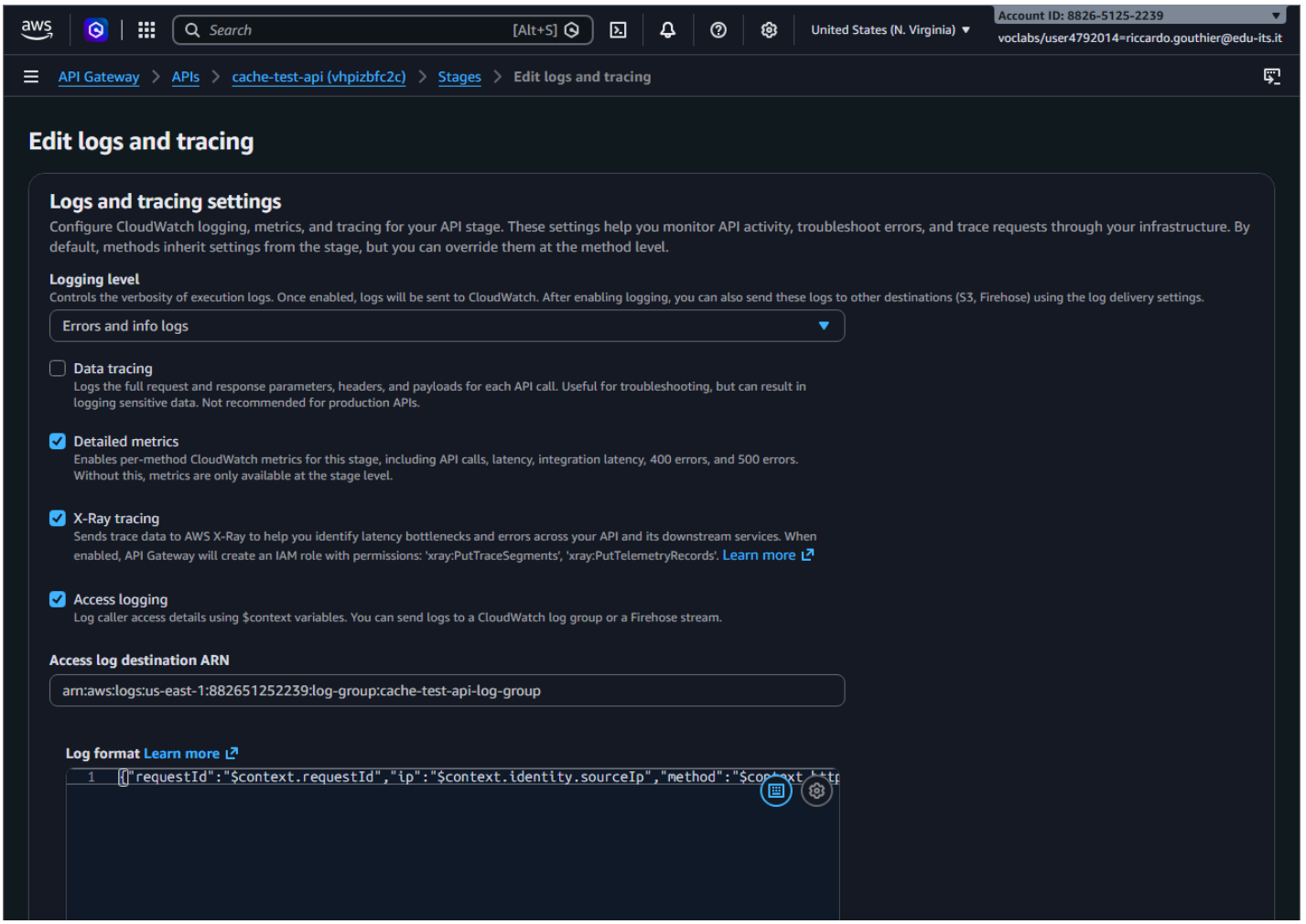
Creare l'infrastruttura per il caching



Abilitare il caching



Log group per cache-test-api: cache-test-api-log-group
Abilitare X-Ray



Abilitare X-Ray x la funzione lambda
~ $ aws lambda update-function-configuration --function-name cache-demo --tracing-config Mode=Active --region us-east-1


```
{
    "FunctionName": "cache-demo",
    "FunctionArn": "arn:aws:lambda:us-east-1:882651252239:function:cache-demo",
    "Runtime": "python3.12",
    "Role": "arn:aws:iam::882651252239:role/LabRole",
    "Handler": "lambda_function.lambda_handler",
    "CodeSize": 437,
    "Description": "",
    "Timeout": 3,
    "MemorySize": 128,
    "LastModified": "2026-10-19T13:44:53.000+0000",
    "CodeSha256": "mXDqDowpc50NgRaSXqGtUkexQ6MvzbABzPXOKgNfyiw=",
    "Version": "$LATEST",
    "TracingConfig": {
        "Mode": "Active"
    },
    "RevisionId": "965a9ec1-b177-416f-8415-7f0c1356d52e",
    "State": "Active",
    "LastUpdateStatus": "InProgress",
```

```
    "LastUpdateStatusReason": "The function is being created.",
    "LastUpdateStatusReasonCode": "Creating",
    "PackageType": "Zip",
    "Architectures": [
        "x86_64"
    ],
    "EphemeralStorage": {
        "Size": 512
    },
    "SnapStart": {
        "ApplyOn": "None",
        "OptimizationStatus": "Off"
    },
    "RuntimeVersionConfig": {
        "RuntimeVersionArn":
"arn:aws:lambda:us-east-1::runtime:36aeb19f3326e129d9f096dea0a3c6a9ca8dce2610e62d2b2060747602d
863ae"
    },
    "LoggingConfig": {
        "LogFormat": "Text",
        "LogGroup": "cache-test-api-log-group"
    }
}
```


```
~ $ aws lambda get-function-configuration --function-name cache-demo --region us-east-1 --query
TracingConfig

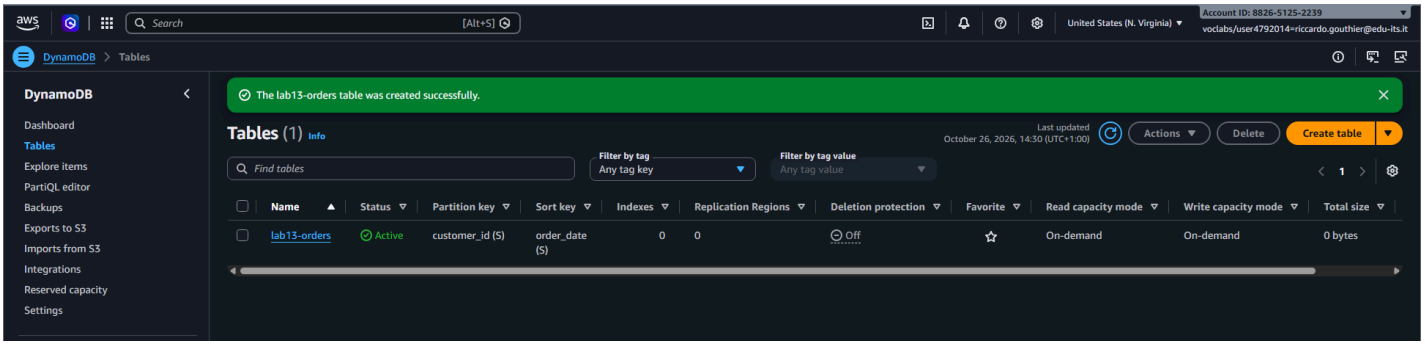
{
    "Mode": "Active"
}
```

Ping alla lambda

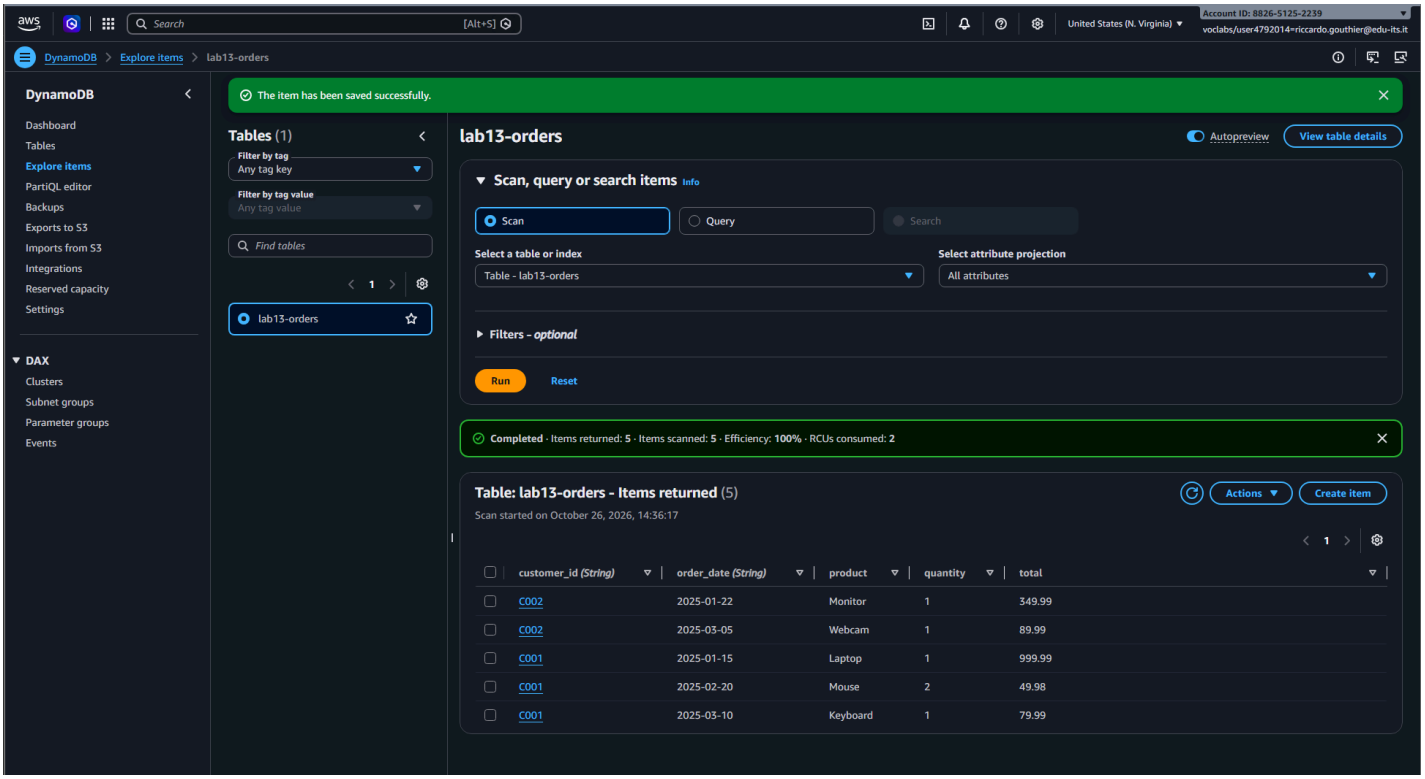
```
~ $ for i in 1 2 3 4 5 6; do curl -s -o /dev/null -w "miss$i: %{http_code} %{time_total}s\n"
"https://vhpizbfc2c.execute-api.us-east-1.amazonaws.com/dev/products?category=xray$i"; done
miss1: 200 0.775138s
miss2: 200 0.542250s
miss3: 200 0.552028s
miss4: 200 0.556631s
miss5: 200 0.537894s
miss6: 200 0.540394s
```

Lab 13

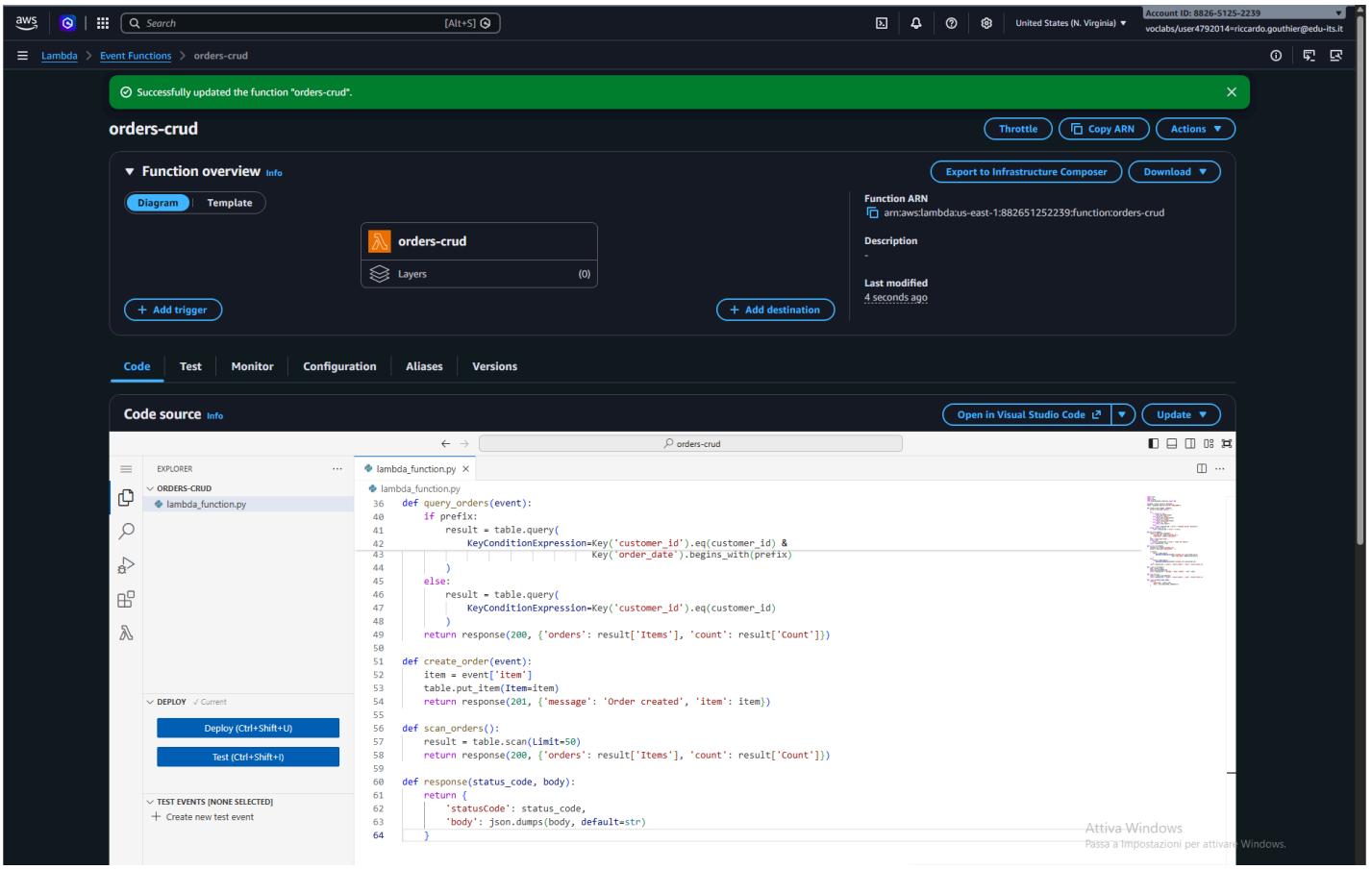
Creare la tabella DynamoDB



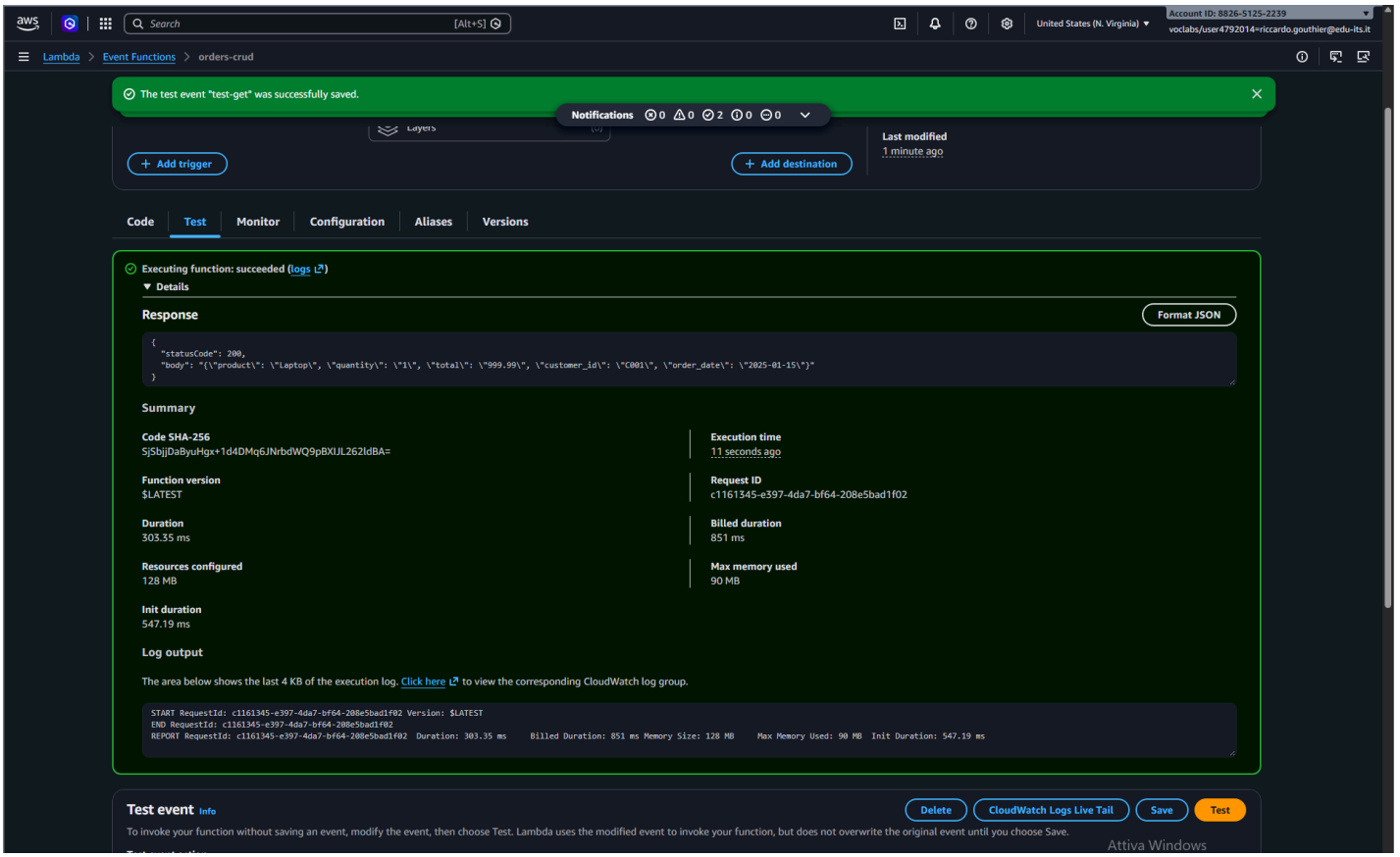
Inserire dati dalla console



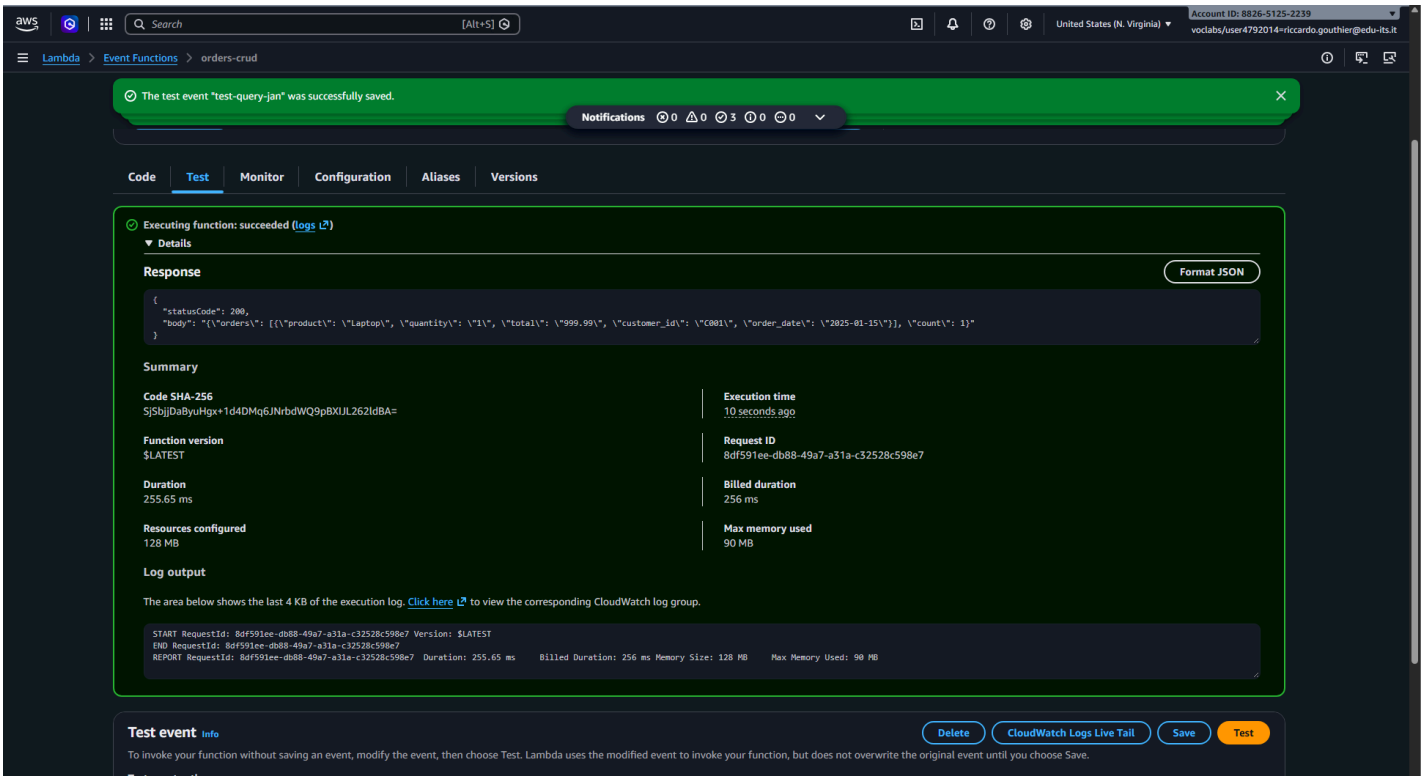
Creare la funzione Lambda CRUD



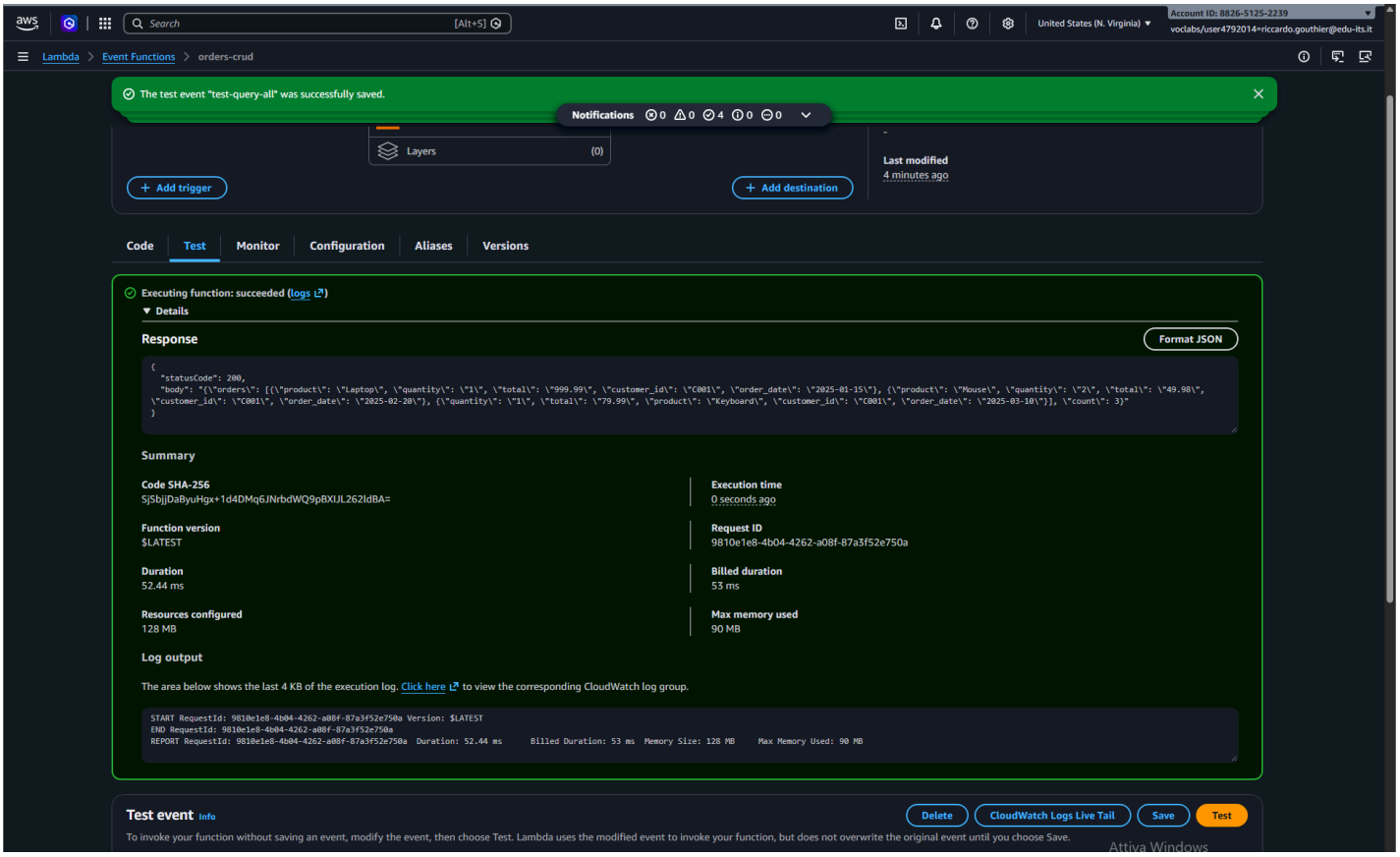
Testare le operazioni GetItem



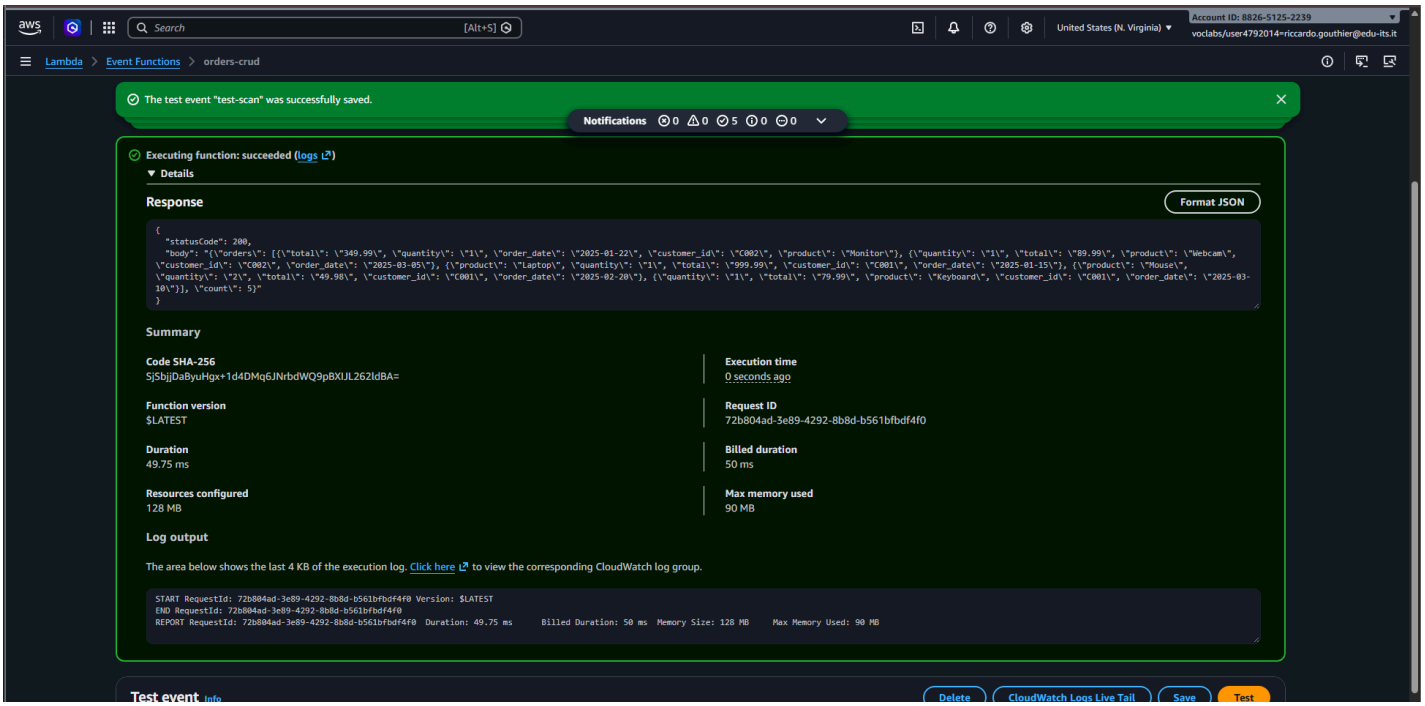
Query gennaio C001



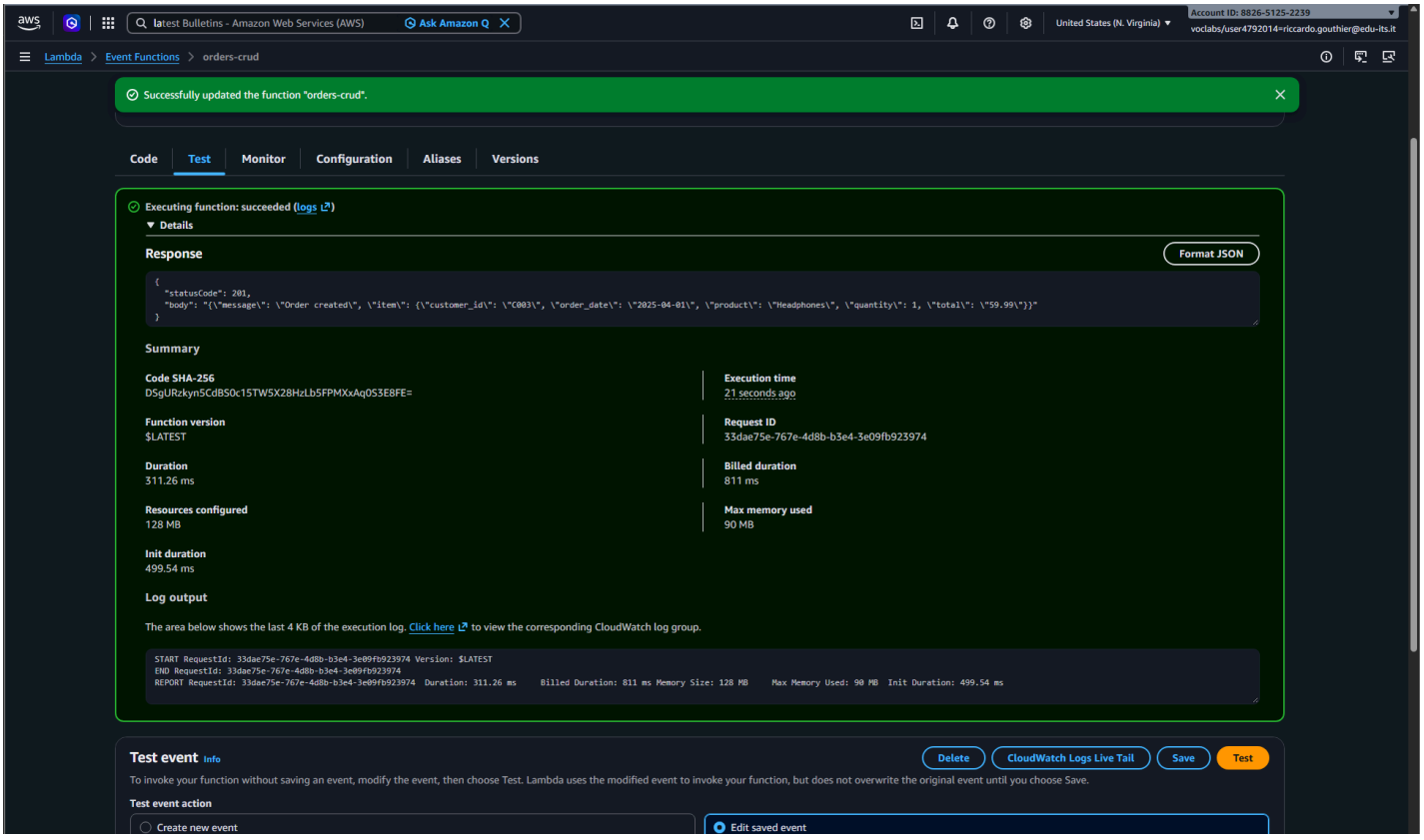
Query tutti C001



Scan



Creare un nuovo ordine



per creare un nuovo ordine il codice non legge i numeri decimali quindi dentro la funzione create_order() va implementato l'utilizzo dei decimal dopo averli importati
modifiche al codice:

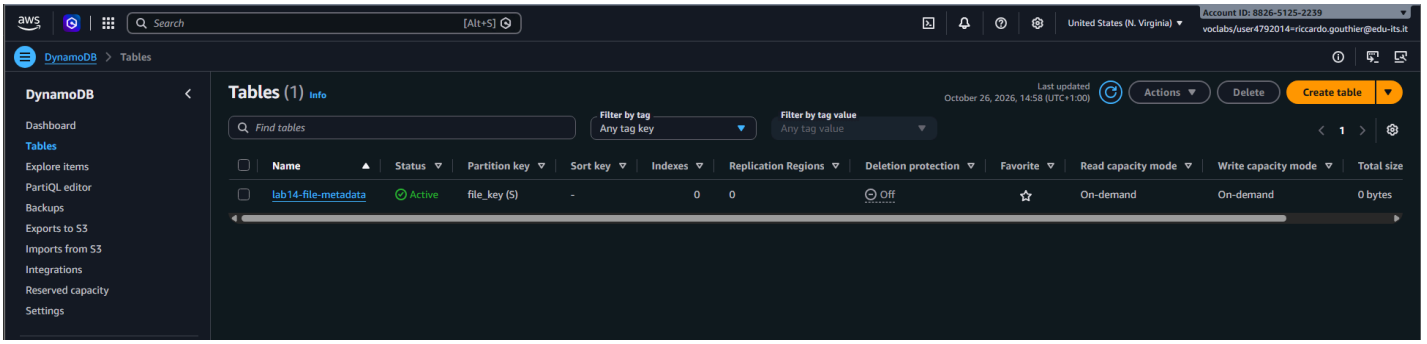
- from decimal import Decimal
- def create_order(event):

item = json.loads(json.dumps(event['item']), parse_float=Decimal

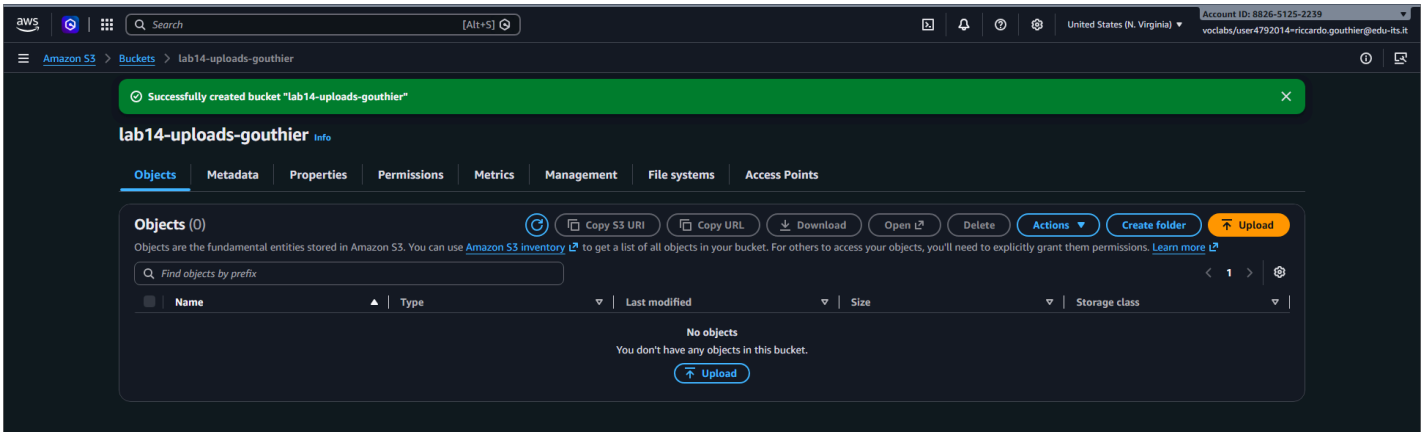
le altre due parti del codice sono tutte corrette anche il resto di create order

Lab 14

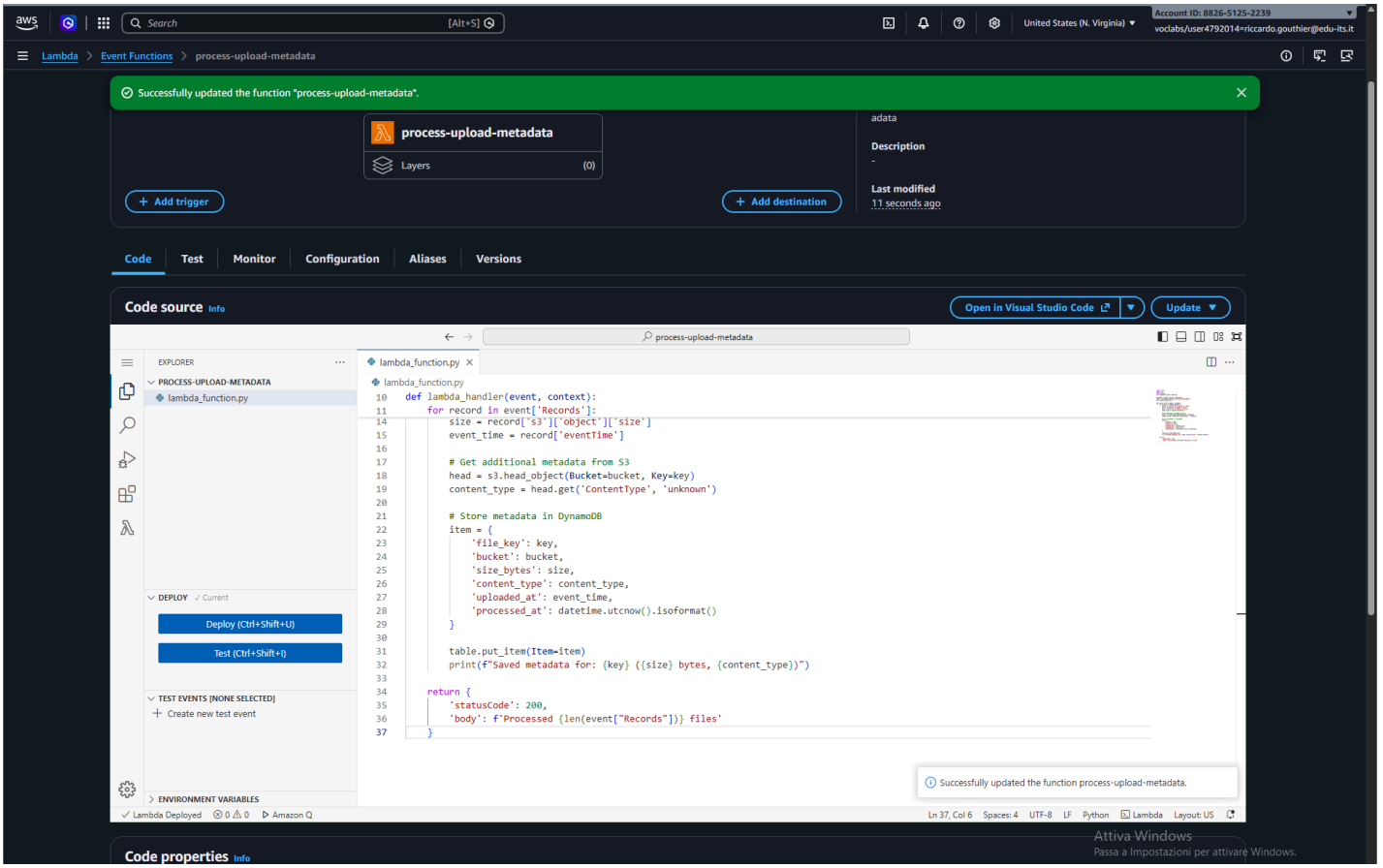
Creare la tabella DynamoDB per i metadati



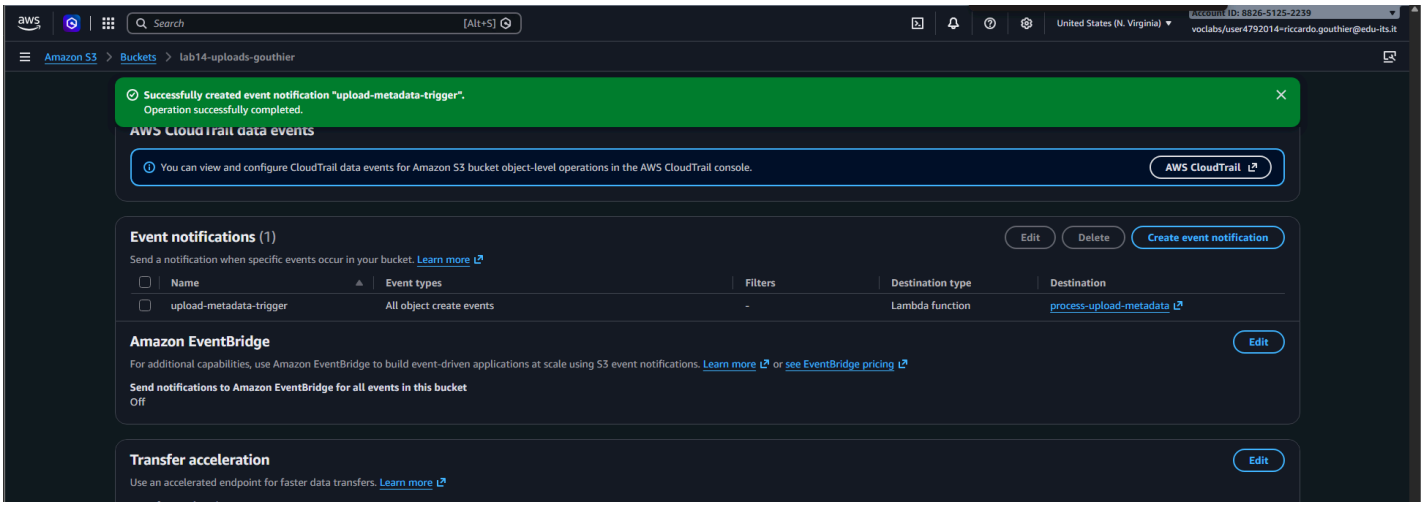
Creare il bucket S3



Creare la funzione Lambda

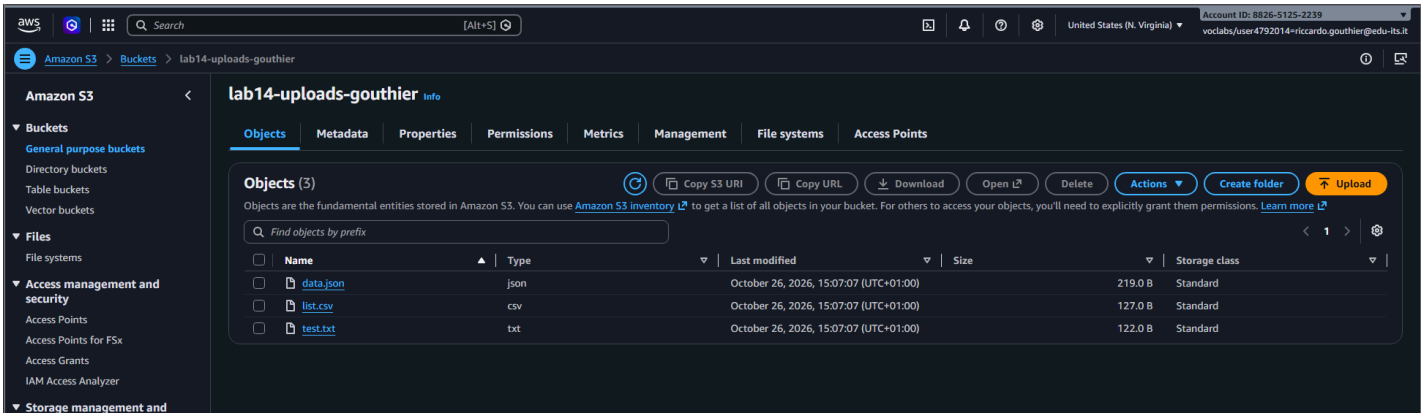


Configurare il trigger S3

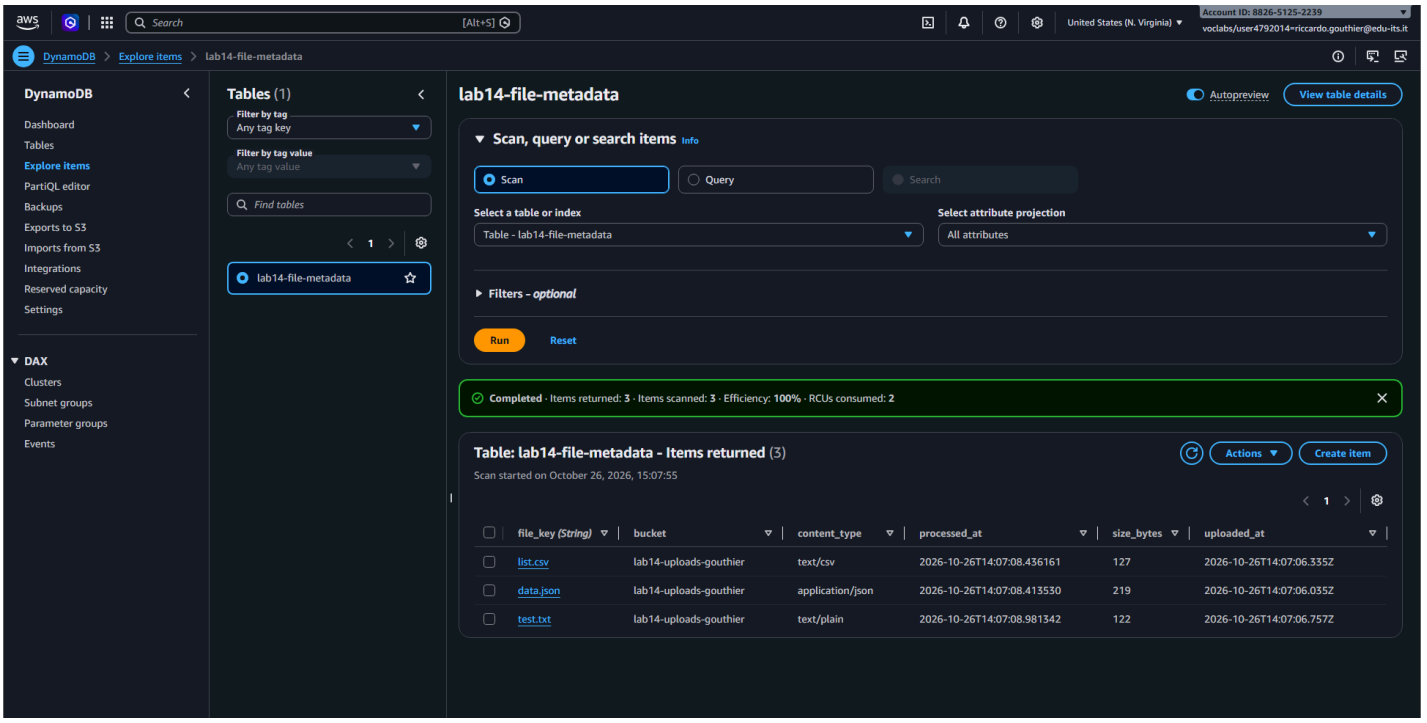


Testare la pipeline

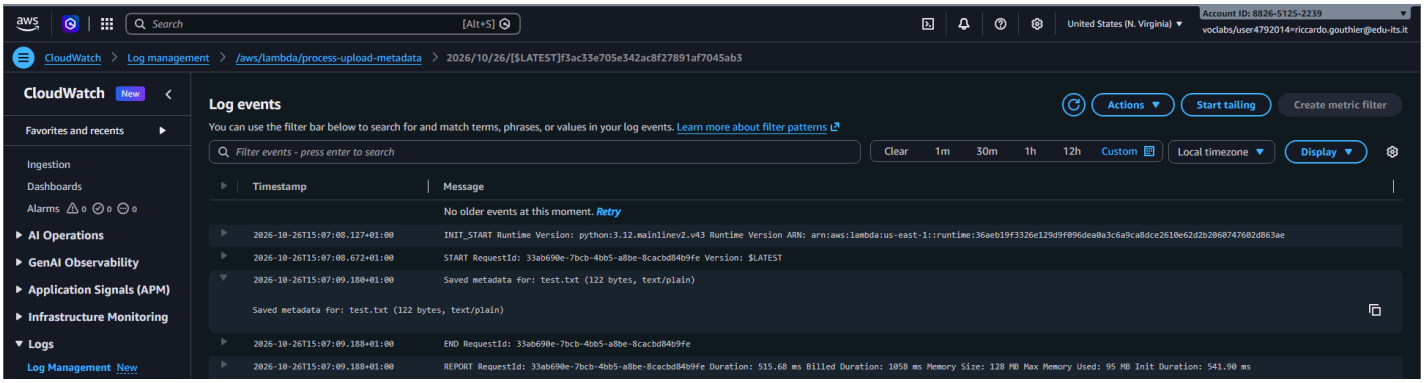
Upload files



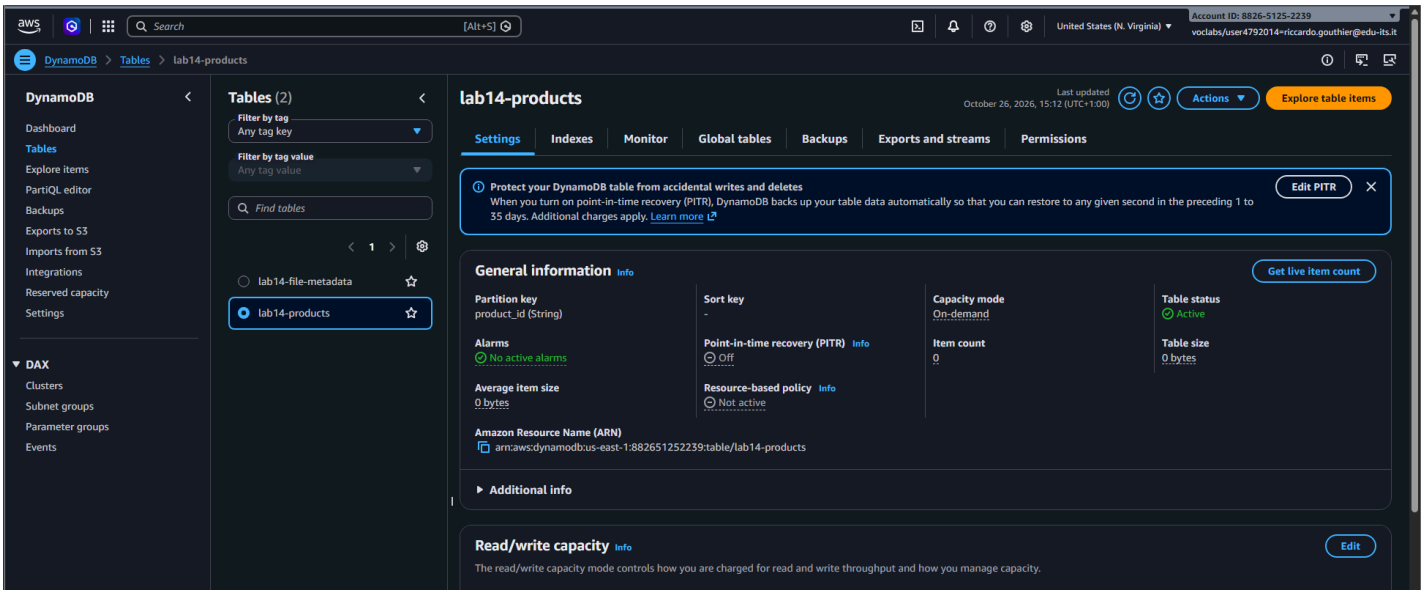
Verifica nella tabella

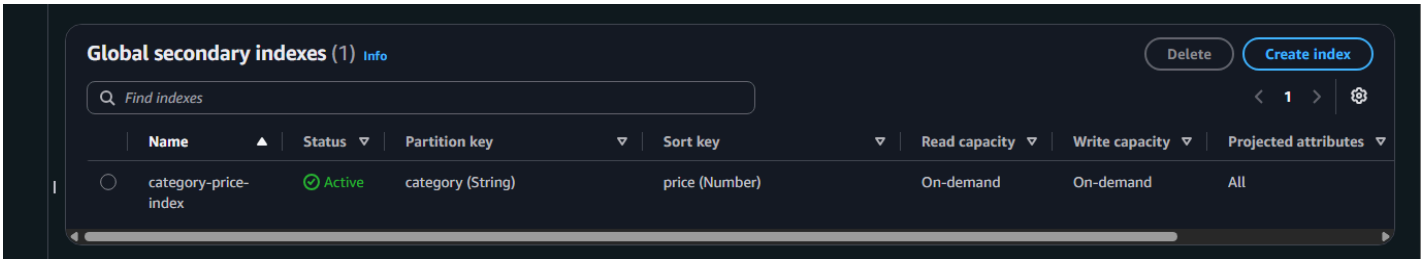


Verificare i log

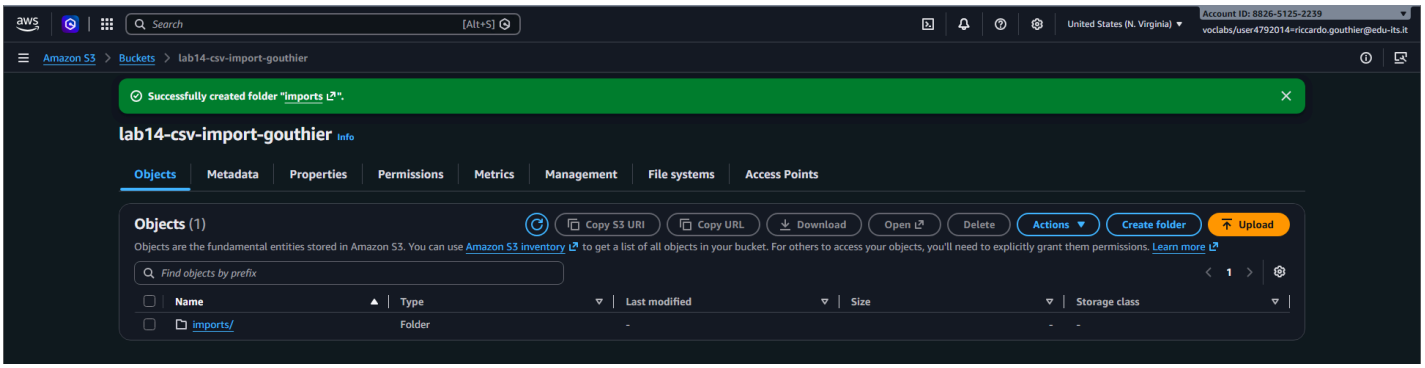


Creare la tabella DynamoDB con GSI

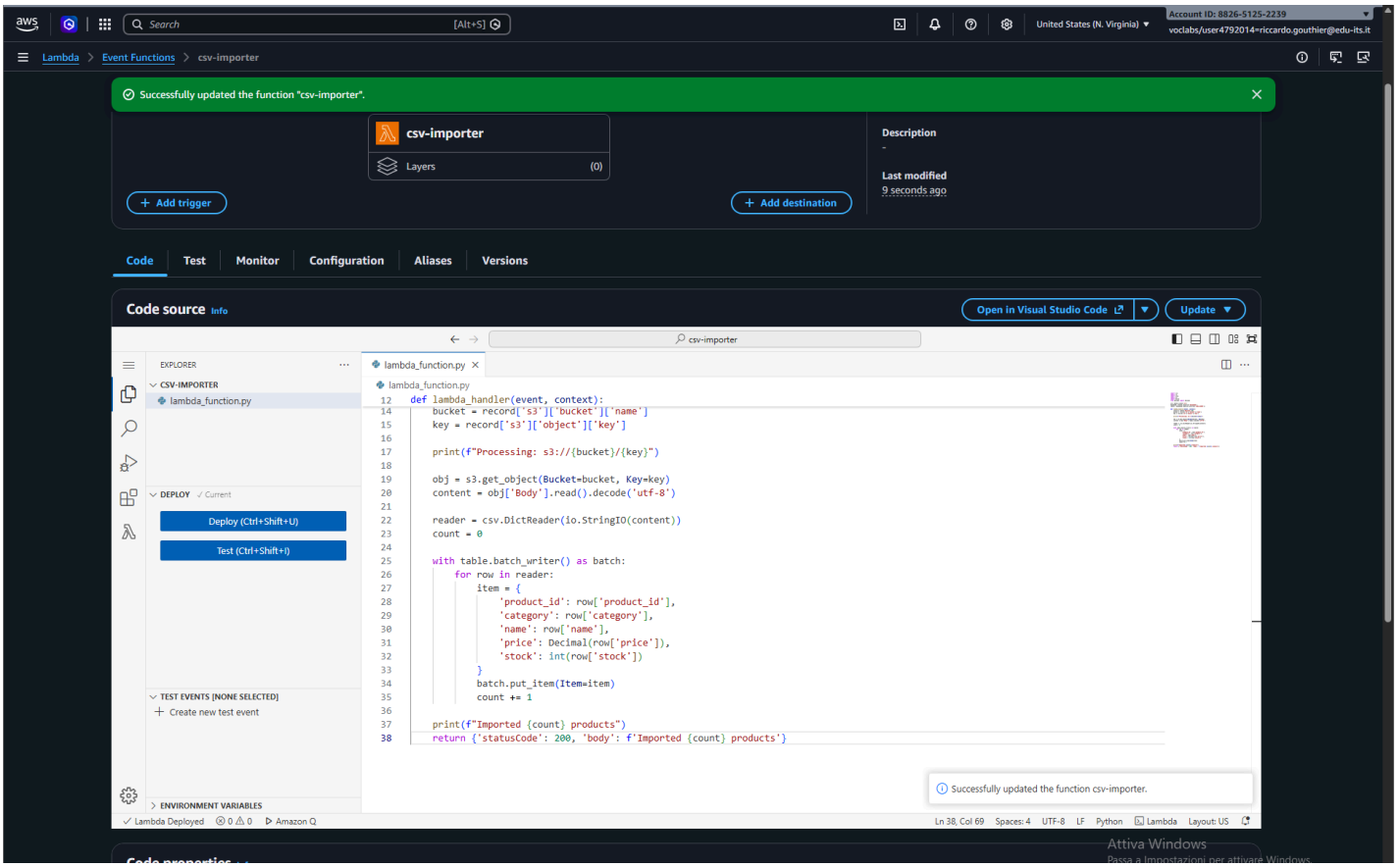




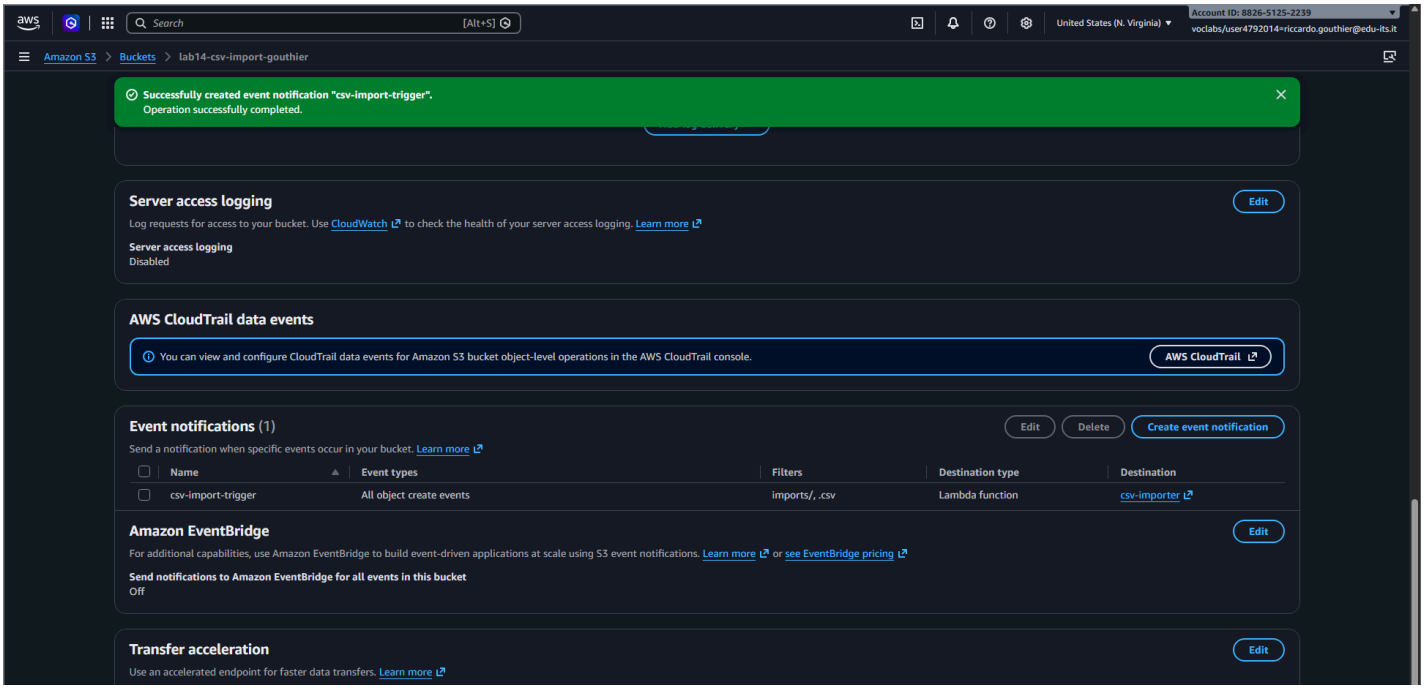
Creare il bucket S3 con folder



Creare la funzione Lambda

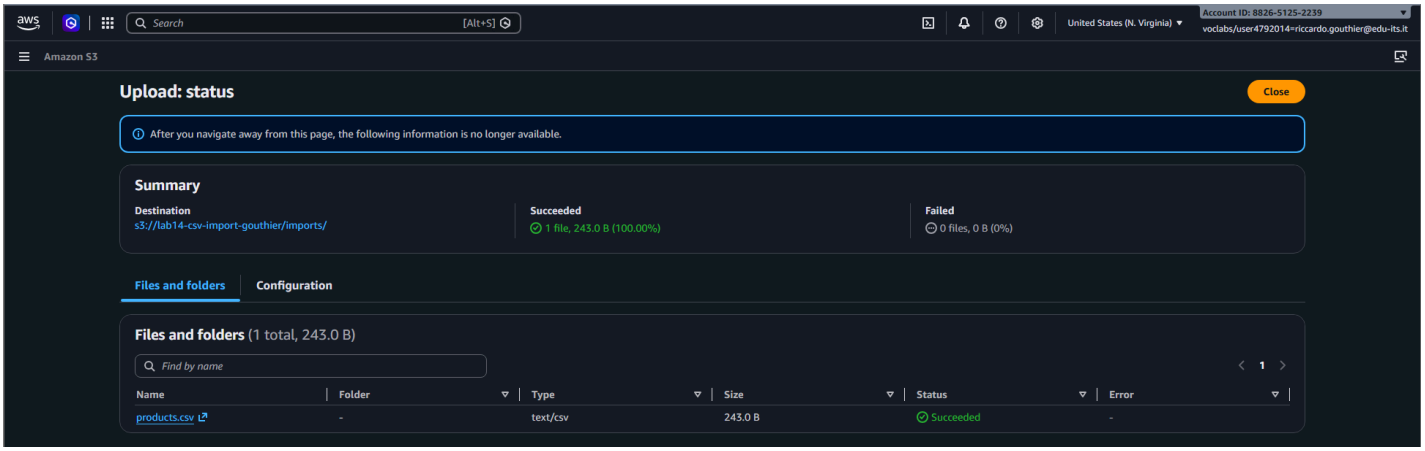


Configurare il trigger con filtro

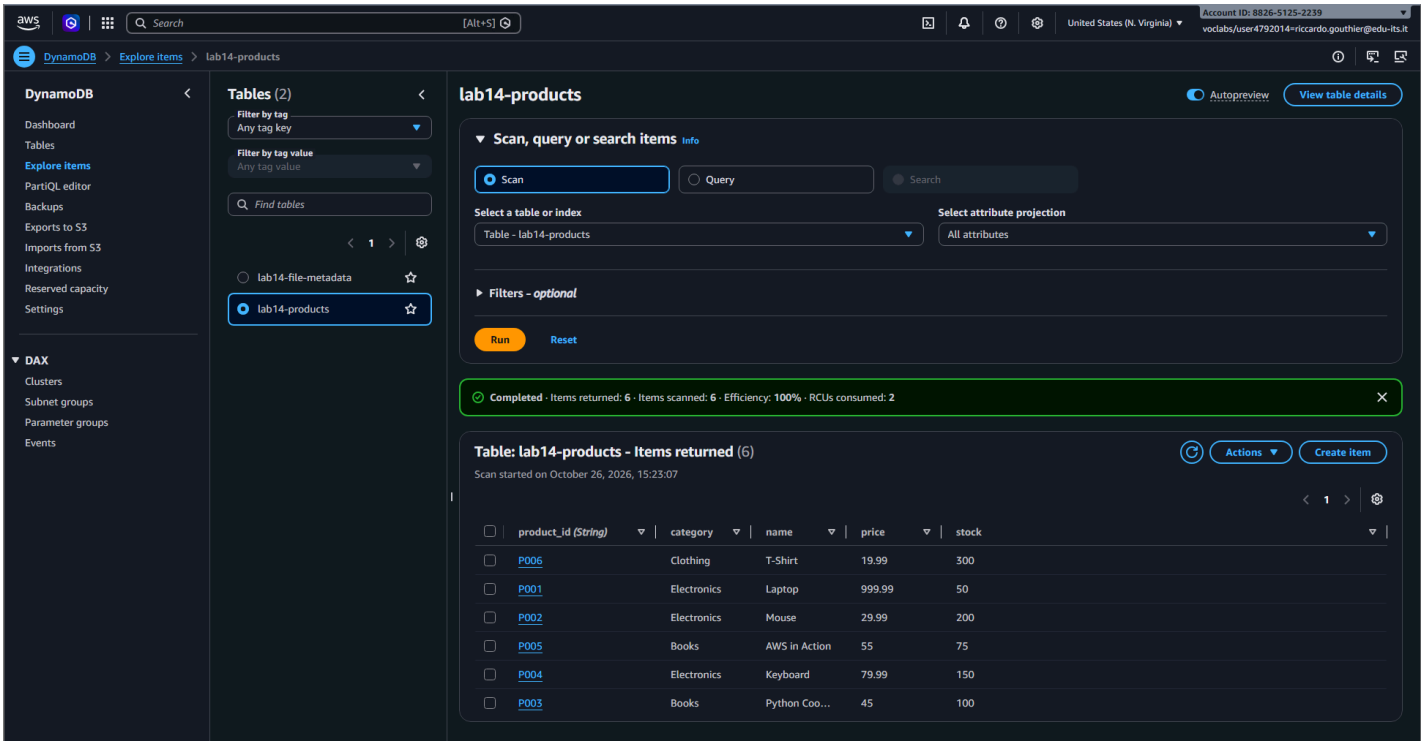


Testare la pipeline

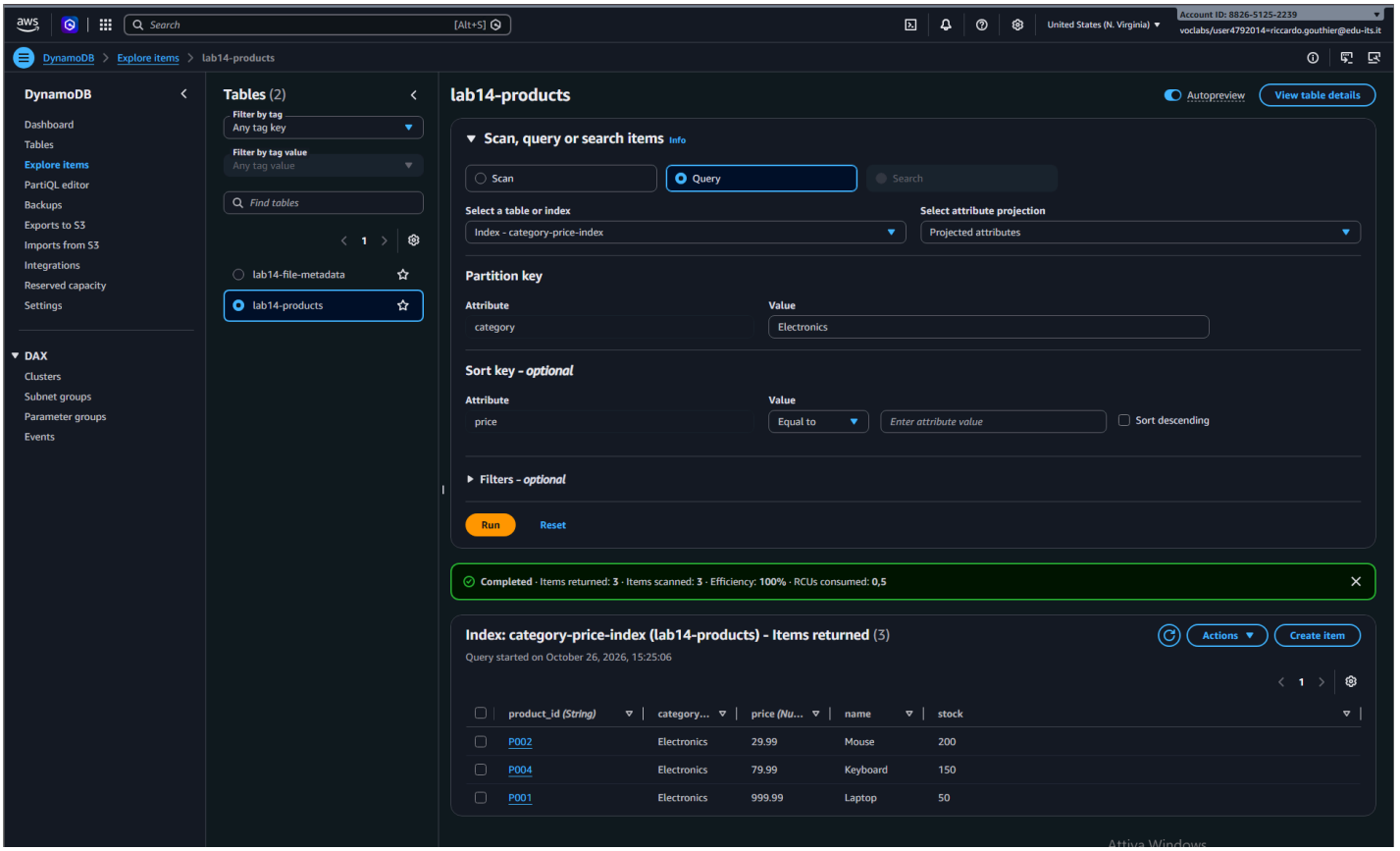
Upload files



Verifica nella tabella



Query con GSI
test con Electronics



test con Books

